

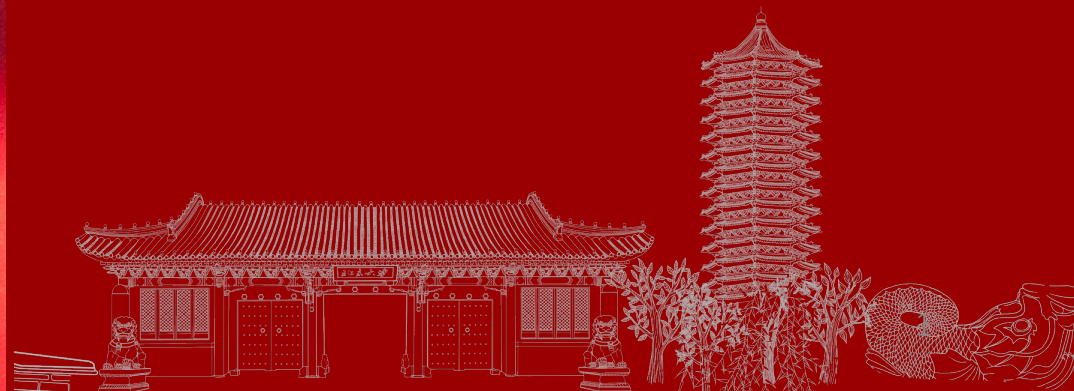
人工智能引论

17. 自然语言处理 -Transformer

授课教师

张辉帅

2026年5月11日



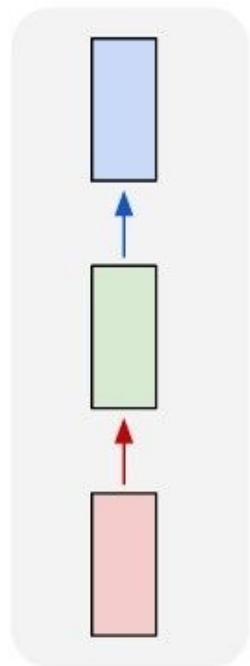
- RNN
- 基于神经网络的自然语言处理方法
- Transformer
- 大语言模型概览

递归神经网络(Recurrent Neural Networks)



北京大学
PEKING UNIVERSITY

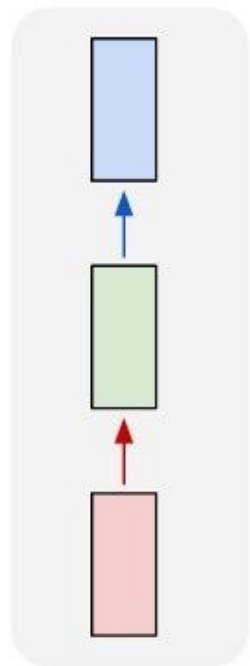
one to one



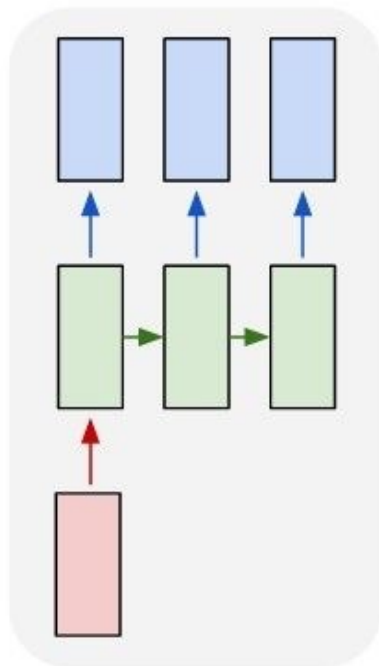
Vanilla Neural Networks

Recurrent Neural Networks

one to one



one to many



例如：图像描述
图像 -> 文字描述



A cat sitting on a suitcase on the floor



A cat is sitting on a tree branch



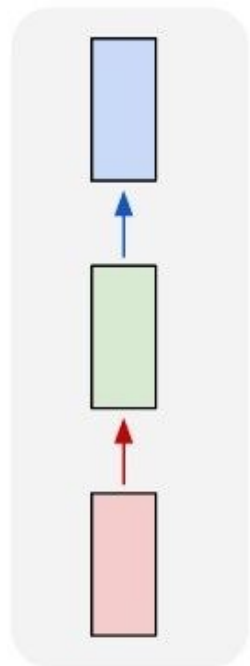
Two people walking on the beach with surfboards



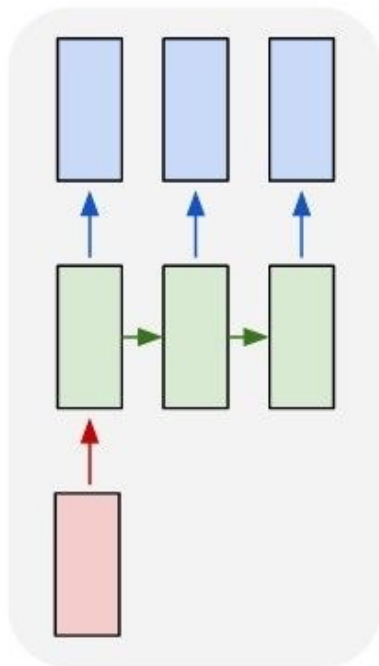
A tennis player in action on the court

Recurrent Neural Networks

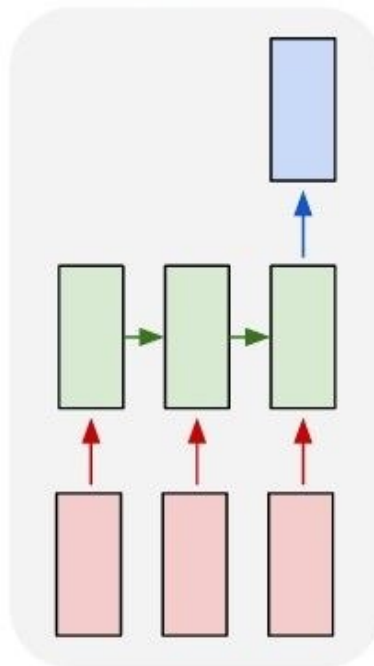
one to one



one to many



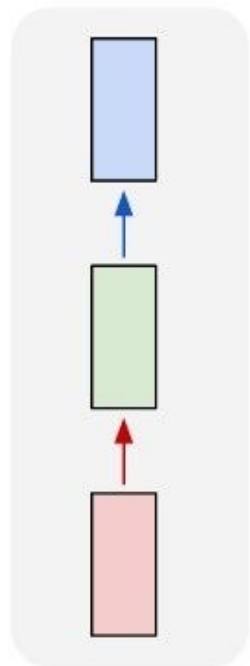
many to one



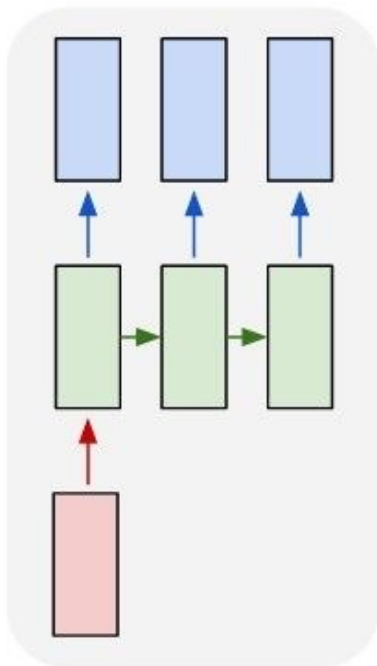
动作预测
视频时间帧 -> 动作类别

Recurrent Neural Networks

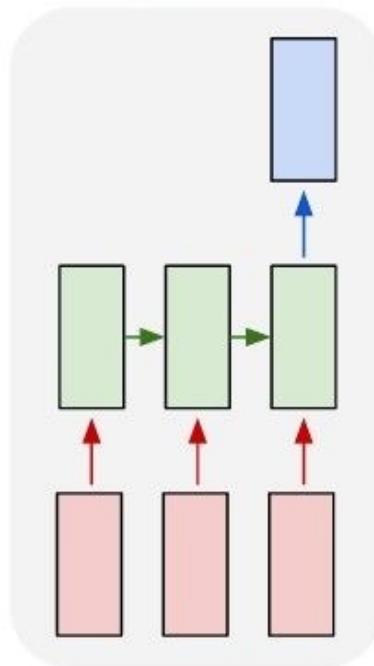
one to one



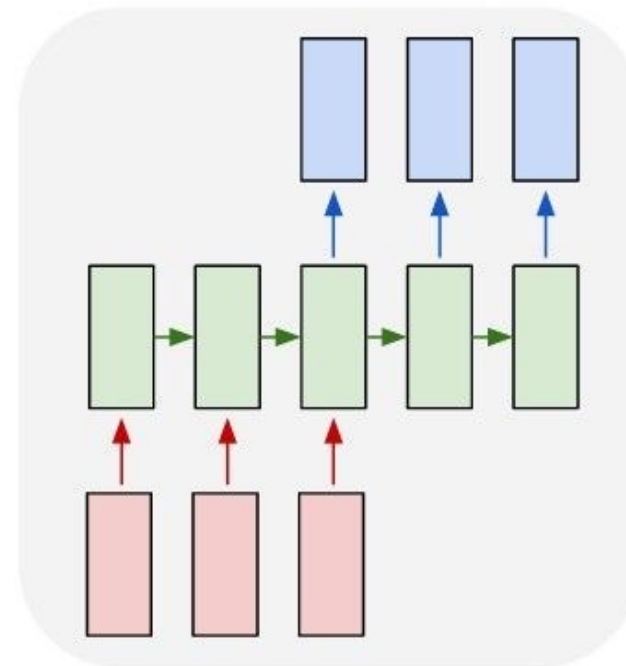
one to many



many to one



many to many



例如：文本翻译
中文句子->英文句子

RNN隐状态更新

循环地使用RNN更新输入x每个时间步的隐状态 h_t :

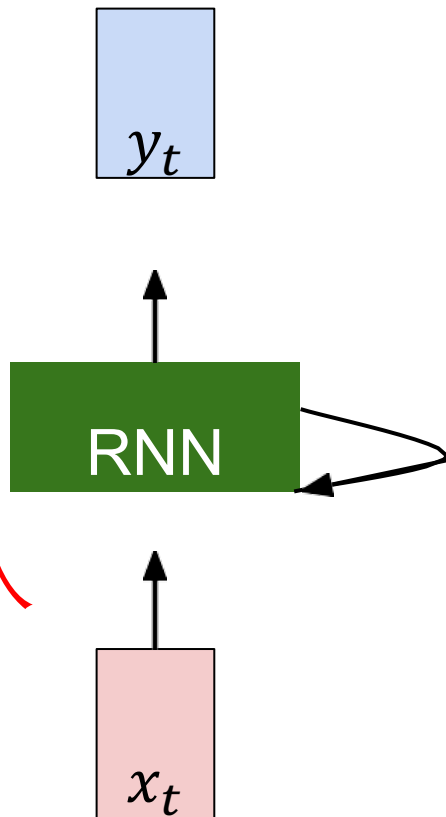
$$\boxed{h_t} = \boxed{f_W}(\boxed{h_{t-1}}, \boxed{x_t})$$

新状态

旧状态

某一时间步的输入

更新状态的函数
(参数为W)



RNN输出预测

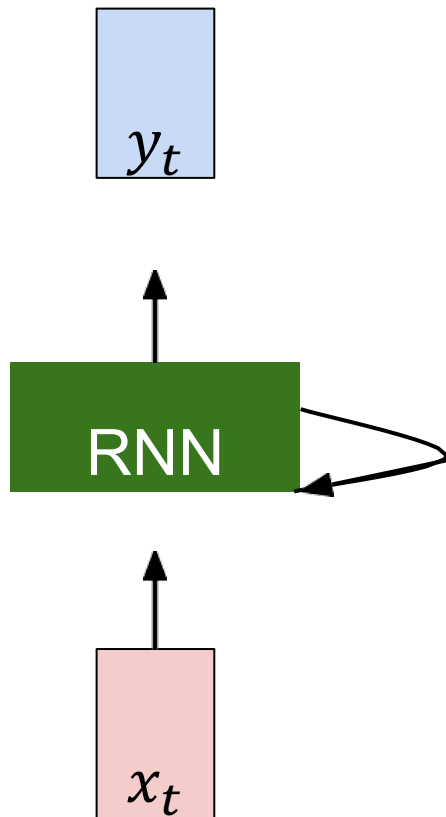
循环地使用RNN更新输入x每个时间步的隐状态 h_t :

$$y_t = f_{W_{hy}}(h_t)$$

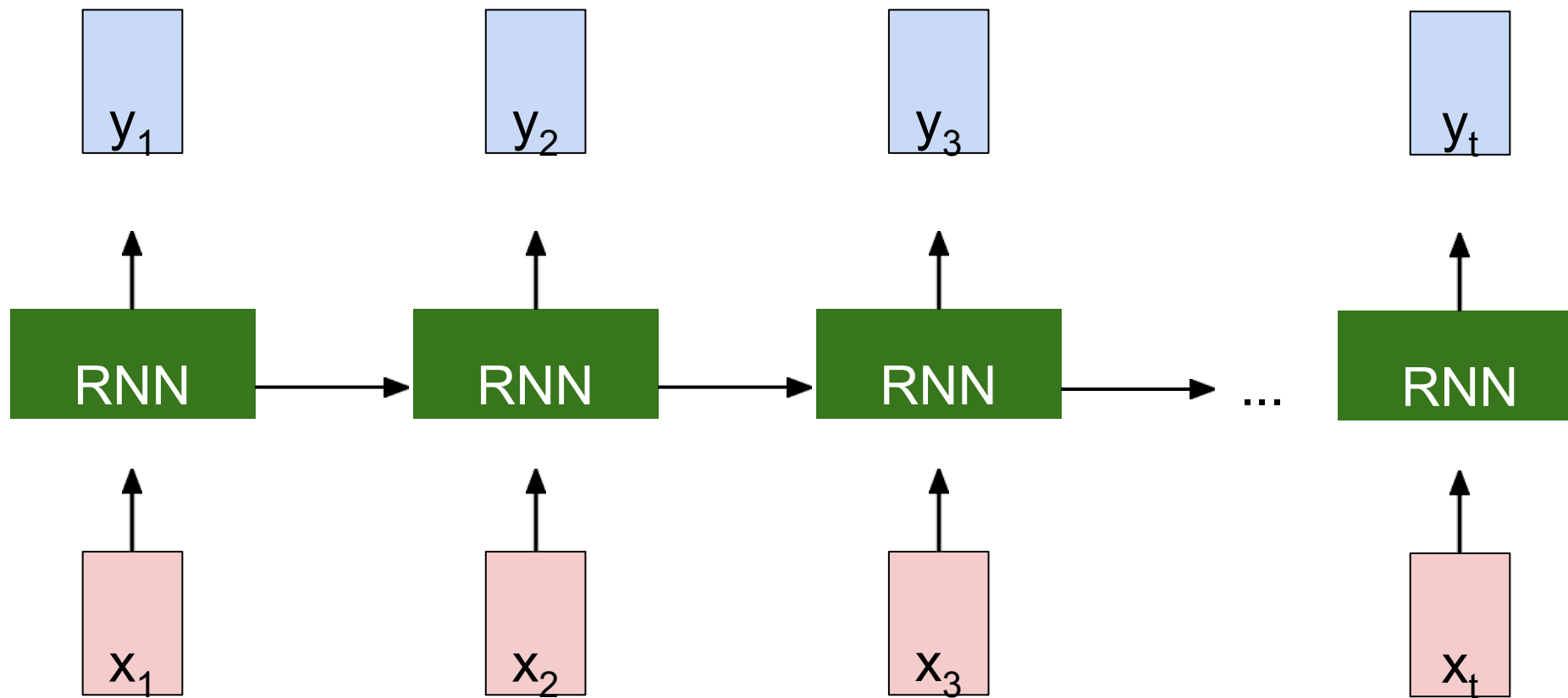
时刻t的预测

时刻t的隐状态

输出预测的函数
(参数为 W_{hy})

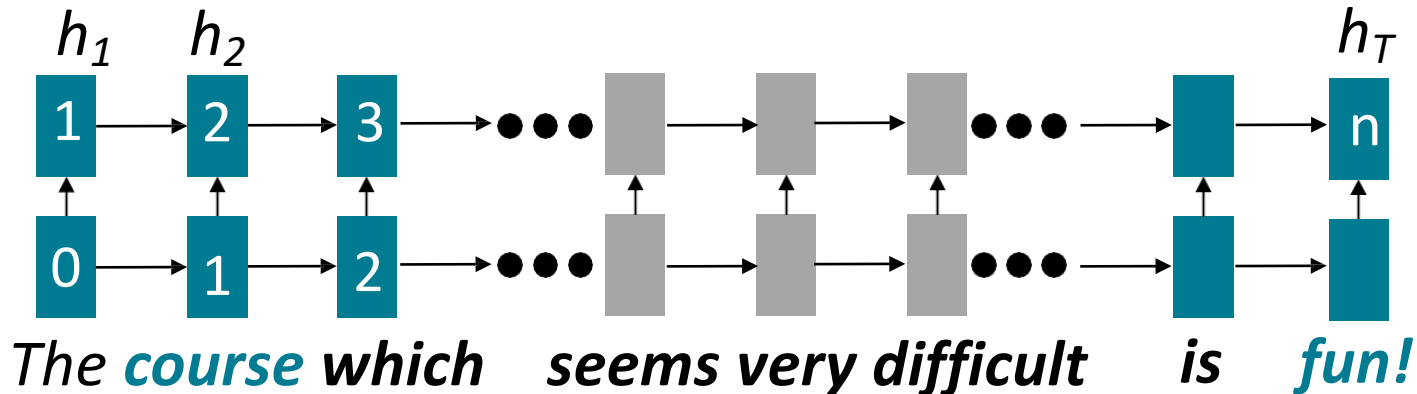


展开 RNN



RNN的优缺点

上面一列是隐状态，下面一列是输入序列。



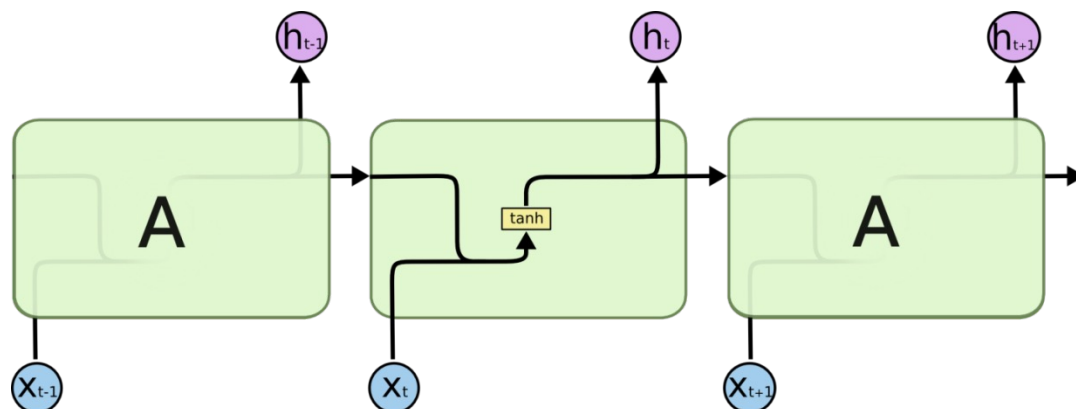
主语(course)和形容词(fun)相隔了许多词，隐状态更新太多次已经遗忘了主语。RNN很难捕获长距离的依赖！

RNN的优缺点

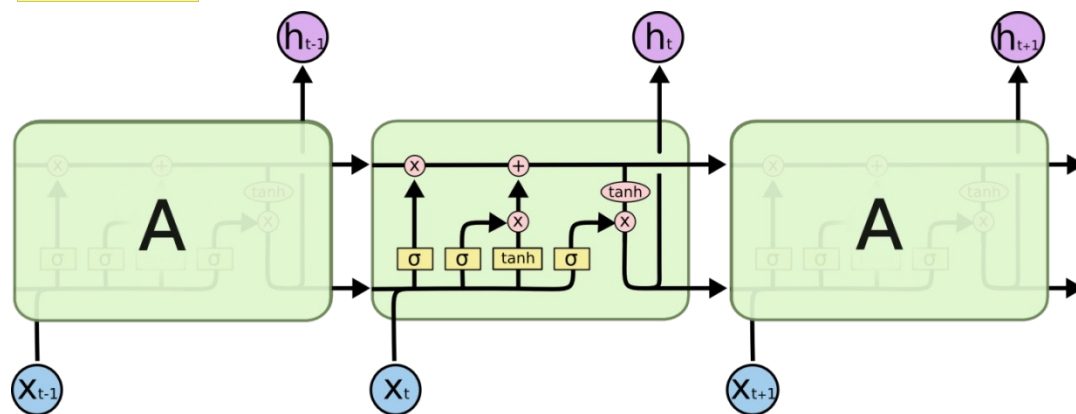
- RNN的优点:
 - 能够处理任意长度数据
 - 输入变长，模型大小不变
- RNN的缺点:
 - 未来状态的计算依赖于过去的状态（无法并行计算）
 - 很难处理长距离依赖（由于梯度衰减问题）
- 后续改进:
 - LSTM（长短期记忆网络）
 - Transformer（SOTA模型）

LSTM & Transformer

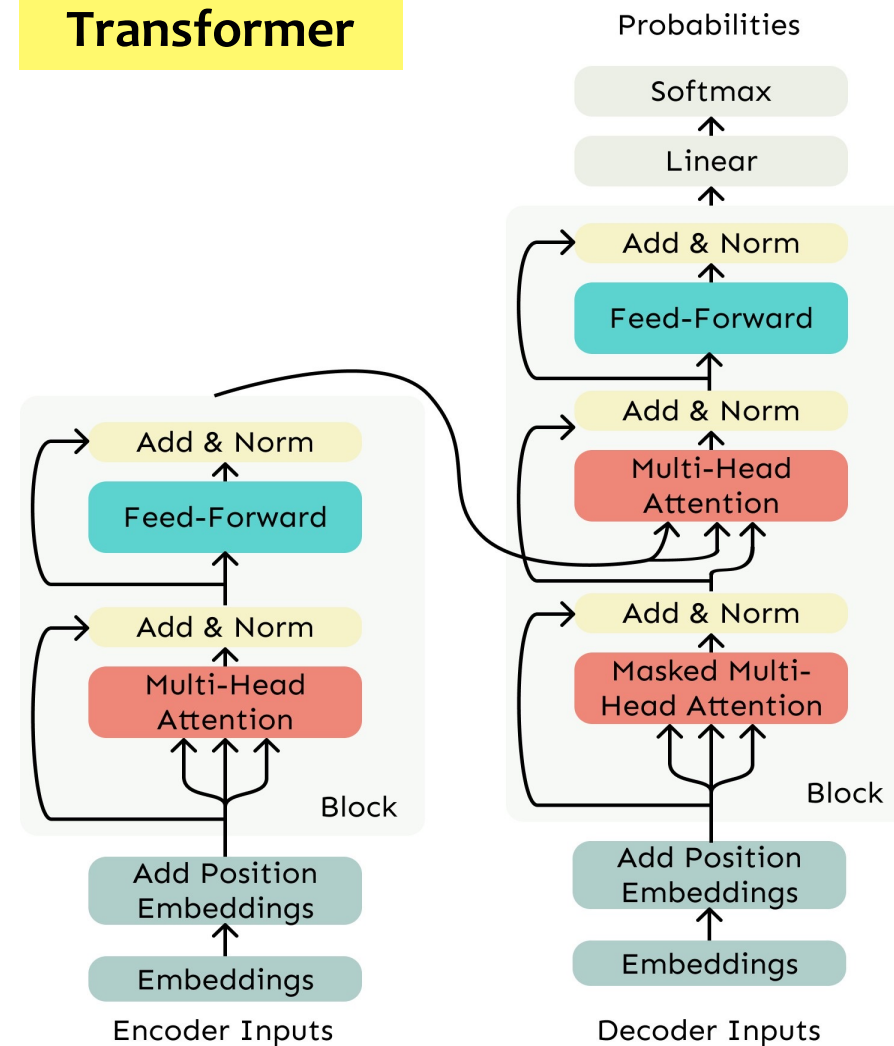
RNNs



LSTM



Transformer



基于神经网络的自然语言处理方法

文本情感分类应用：基于人工设计文本特征

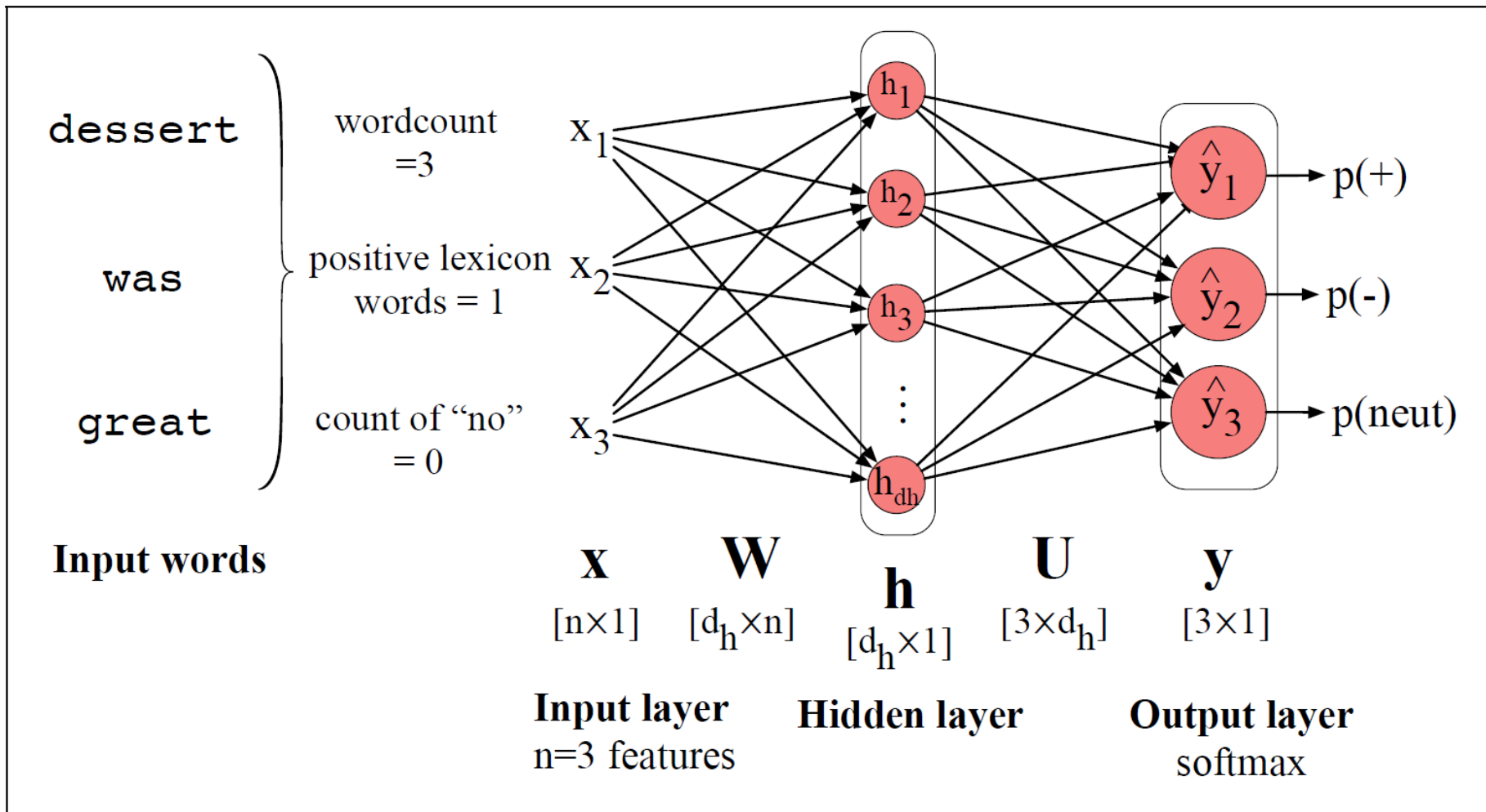
- x_i 是人工设计特征，可以是标量

$$\mathbf{x} = [x_1, x_2, \dots, x_N]$$

$$\mathbf{h} = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b})$$

$$\mathbf{z} = \mathbf{U}\mathbf{h}$$

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{z})$$



基于神经网络的自然语言处理方法

文本情感分类应用：基于word2vec/GloVe的文本特征

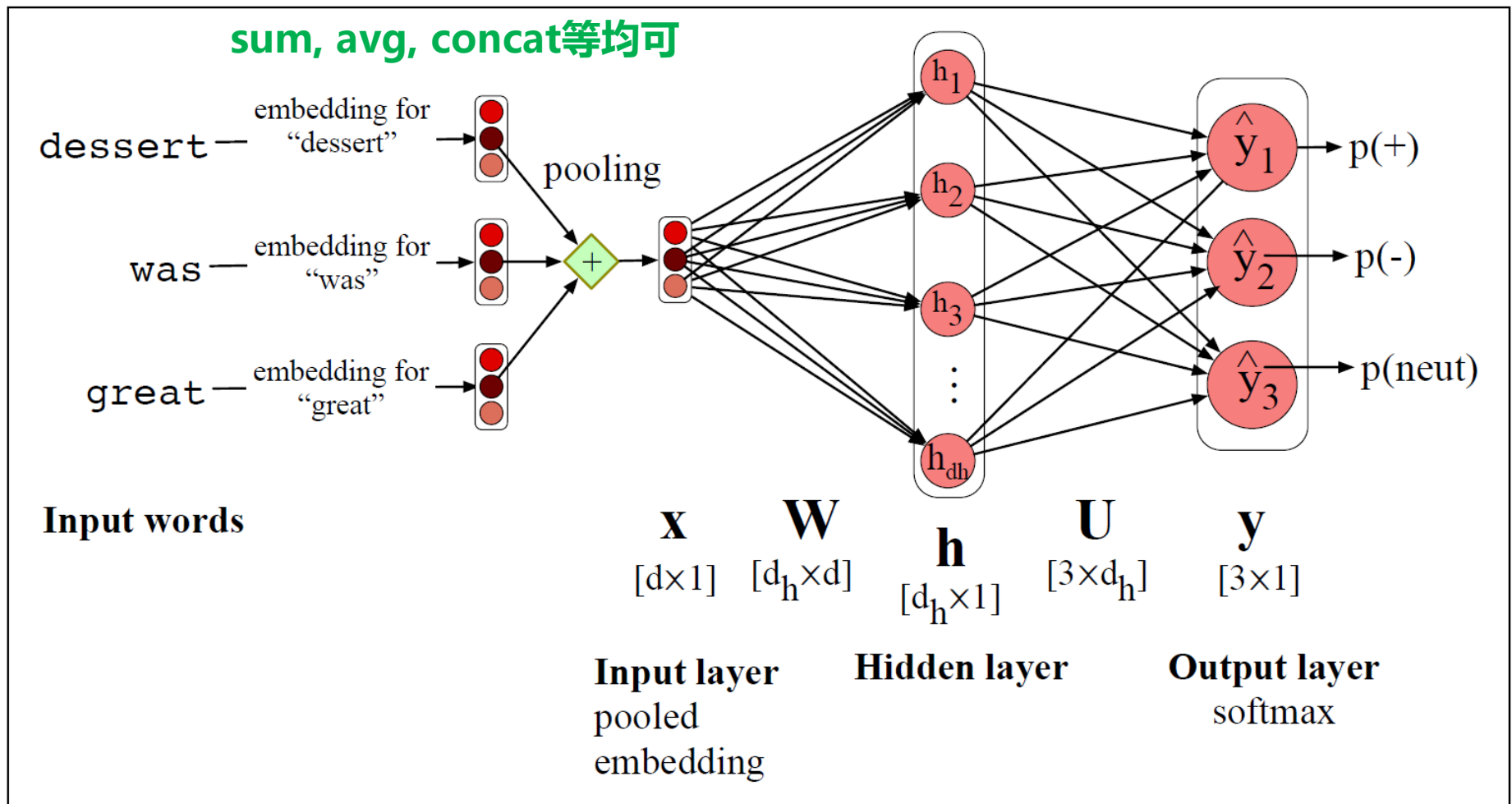
- 文本包含 n 个词 w_i
- $e(w_i)$: word2vec词嵌入

$$\mathbf{x} = \frac{1}{n} \sum_{i=1}^n \mathbf{e}(w_i)$$

$$\mathbf{h} = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b})$$

$$\mathbf{z} = \mathbf{U}\mathbf{h}$$

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{z})$$



基于神经网络的自然语言处理方法

前馈神经语言模型：基于前词上下文预测接下来的词

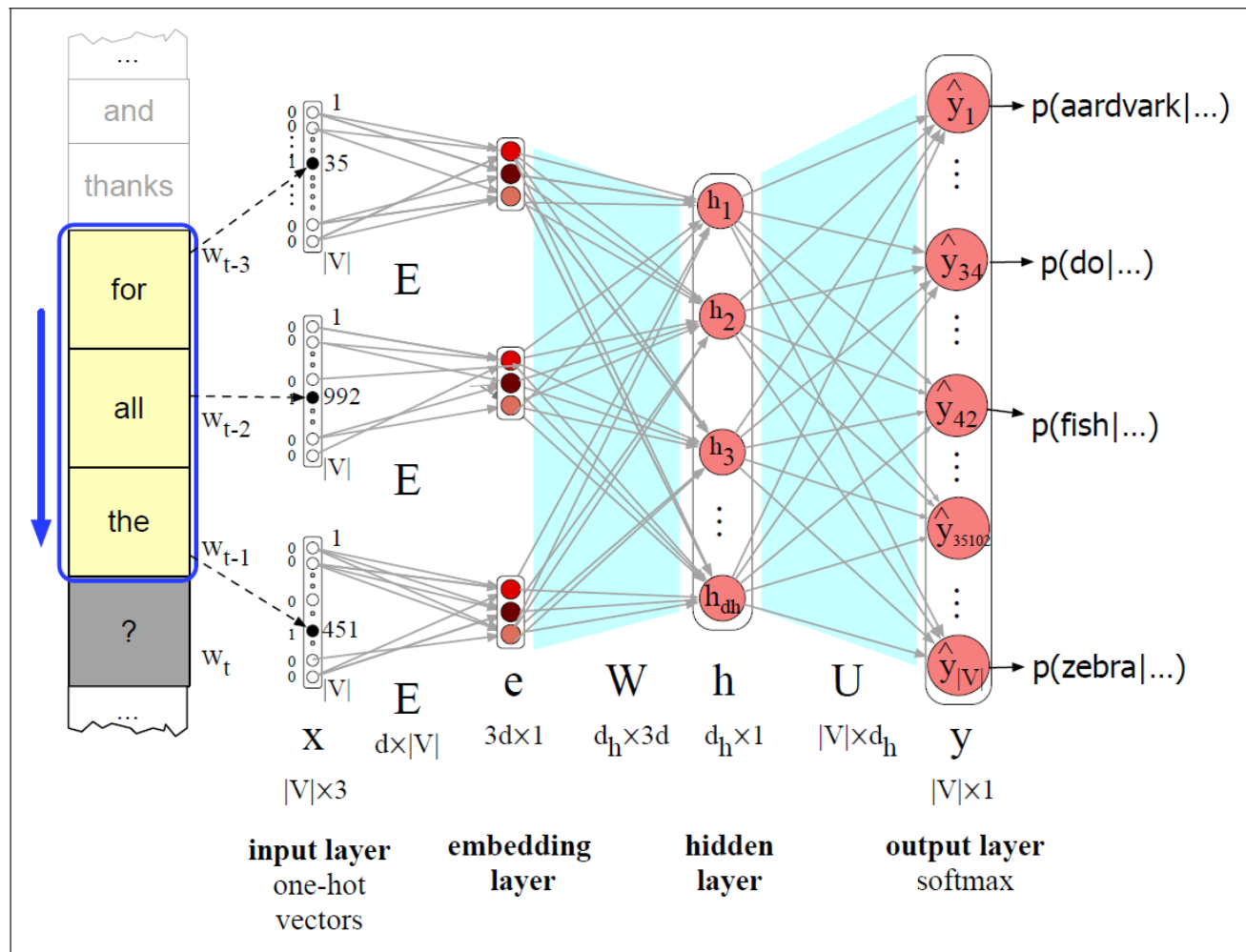
- 输入独热向量
- 给定窗口大小(如3)

$$\mathbf{e} = [\mathbf{E}\mathbf{x}_{t-3}; \mathbf{E}\mathbf{x}_{t-2}; \mathbf{E}\mathbf{x}_{t-1}]$$

$$\mathbf{h} = \sigma(\mathbf{W}\mathbf{e} + \mathbf{b})$$

$$\mathbf{z} = \mathbf{U}\mathbf{h}$$

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{z})$$



基于神经网络的自然语言处理方法

前馈神经语言模型：基于前词上下文预测接下来的词

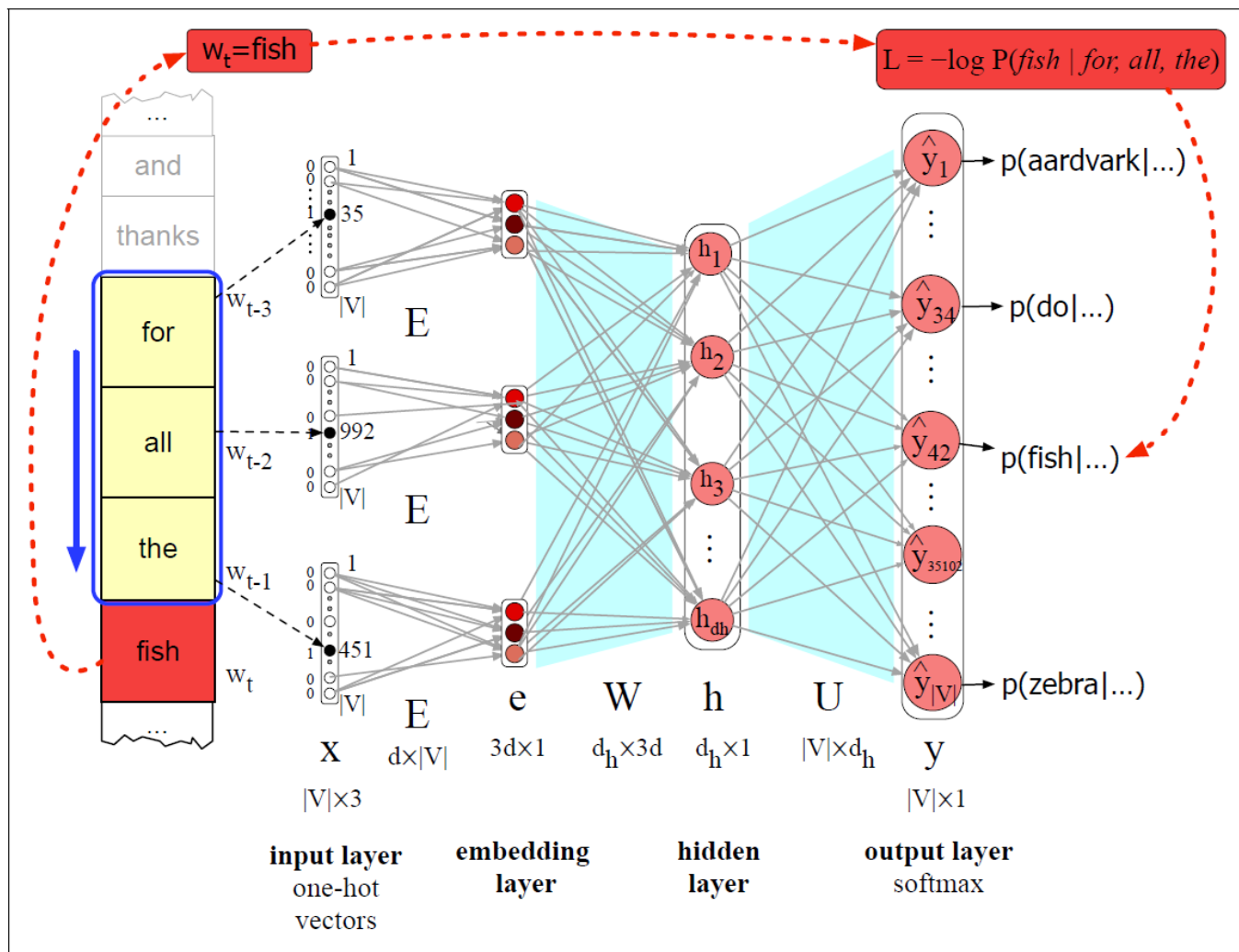
- 输入独热向量
- 给定窗口大小(如3)

$$\mathbf{e} = [\mathbf{E}\mathbf{x}_{t-3}; \mathbf{E}\mathbf{x}_{t-2}; \mathbf{E}\mathbf{x}_{t-1}]$$

$$\mathbf{h} = \sigma(\mathbf{W}\mathbf{e} + \mathbf{b})$$

$$\mathbf{z} = \mathbf{U}\mathbf{h}$$

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{z})$$



基于神经网络的自然语言处理方法

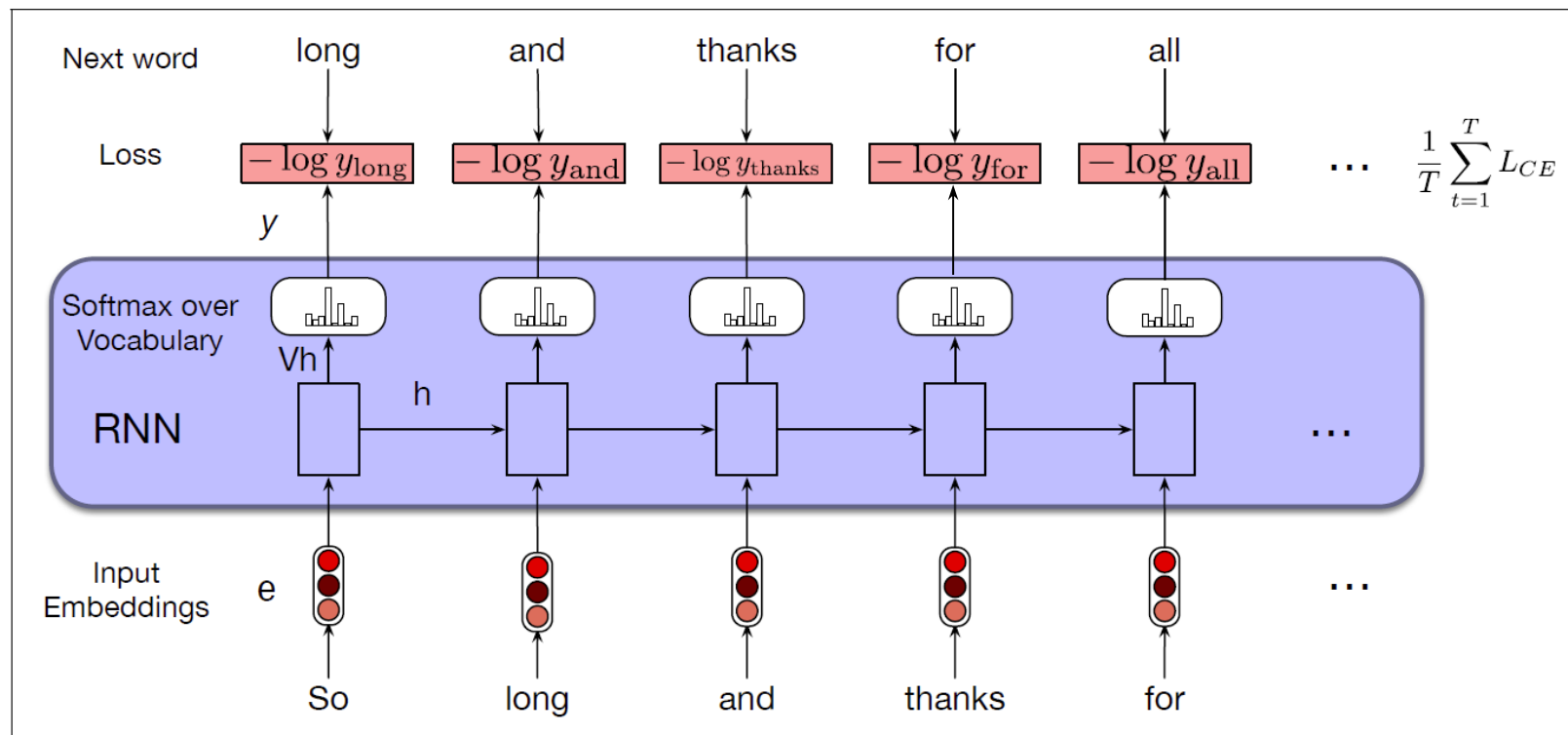
递归神经语言模型：基于前词上下文预测接下来的词

- 输入独热向量
- RNN递归生成
- 无窗口限制

$$\mathbf{e}_t = \mathbf{E}\mathbf{x}_t$$

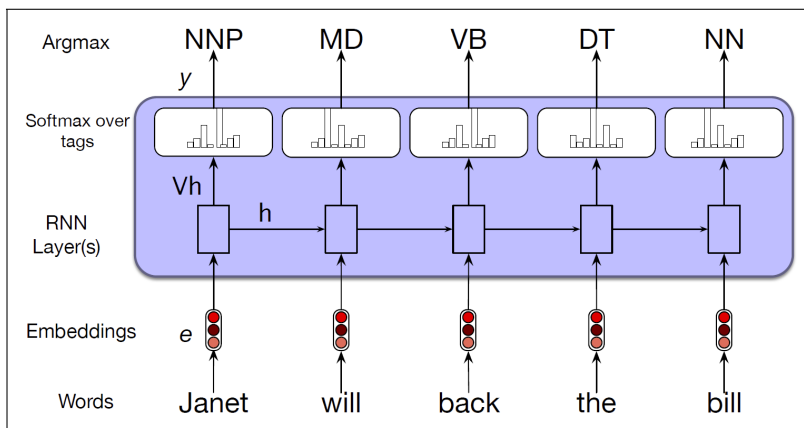
$$\mathbf{h}_t = g(\mathbf{U}\mathbf{h}_{t-1} + \mathbf{W}\mathbf{e}_t)$$

$$\mathbf{y}_t = \text{softmax}(\mathbf{V}\mathbf{h}_t)$$

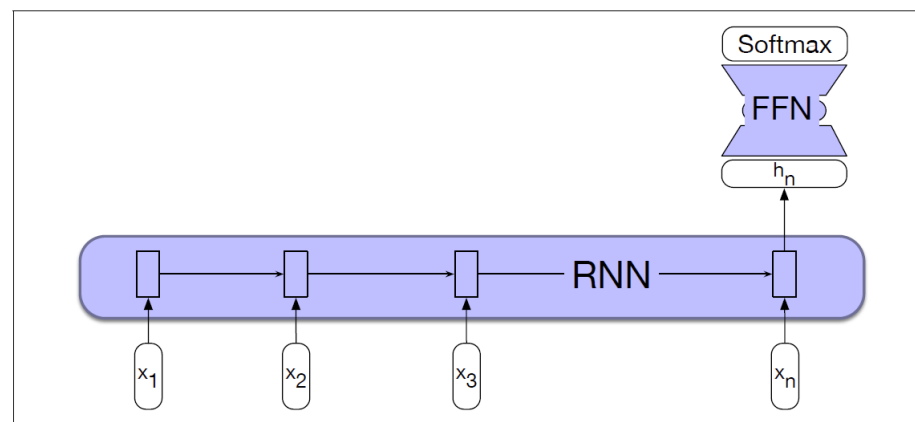


基于神经网络的自然语言处理方法

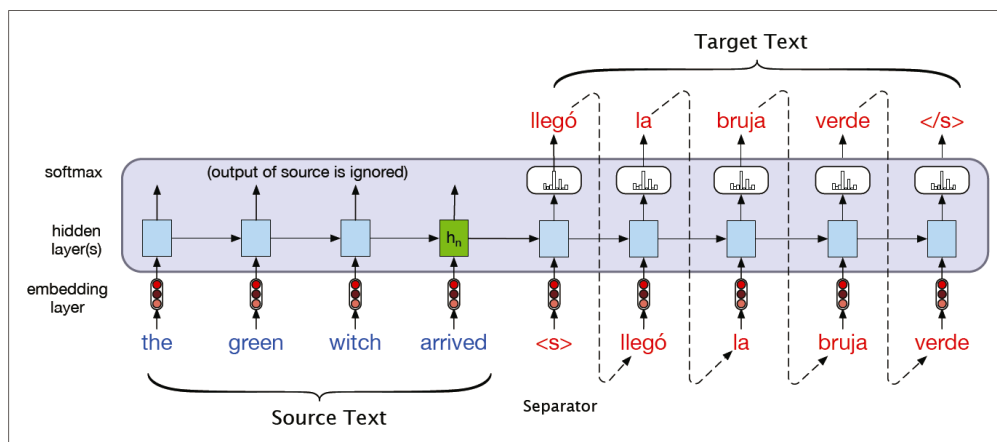
递归神经网络RNN在NLP其他任务中的应用：



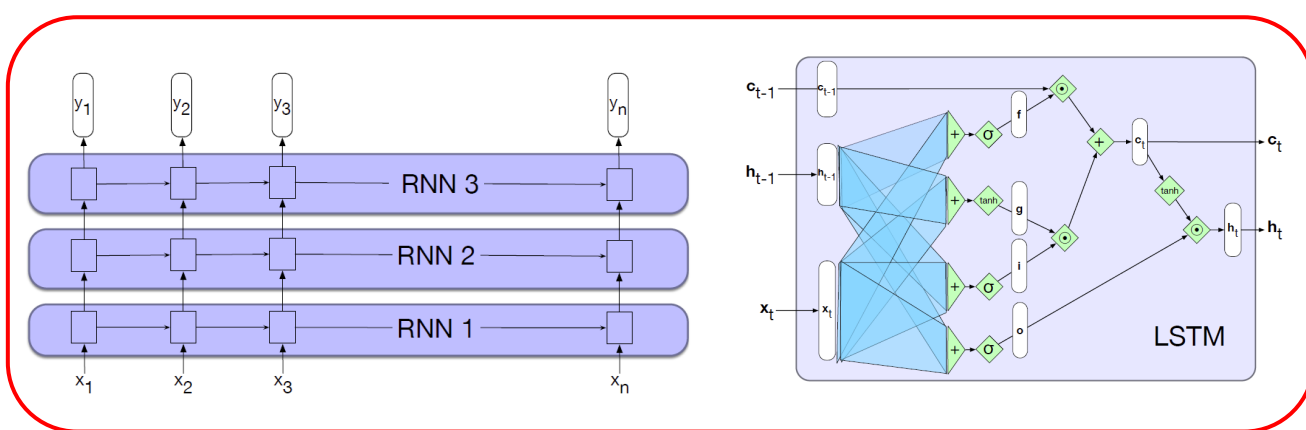
序列标注：例如词性标注



序列分类：例如情感分类、垃圾邮件分类等



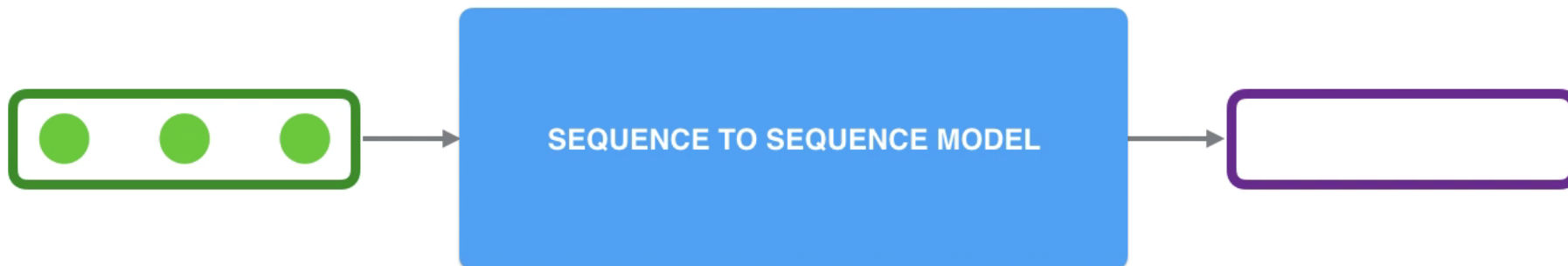
文本生成：例如机器翻译、文本摘要、问答等



改进：多层堆叠RNN、LSTM等...

Transformer

- 序列到序列变换(seq2seq)

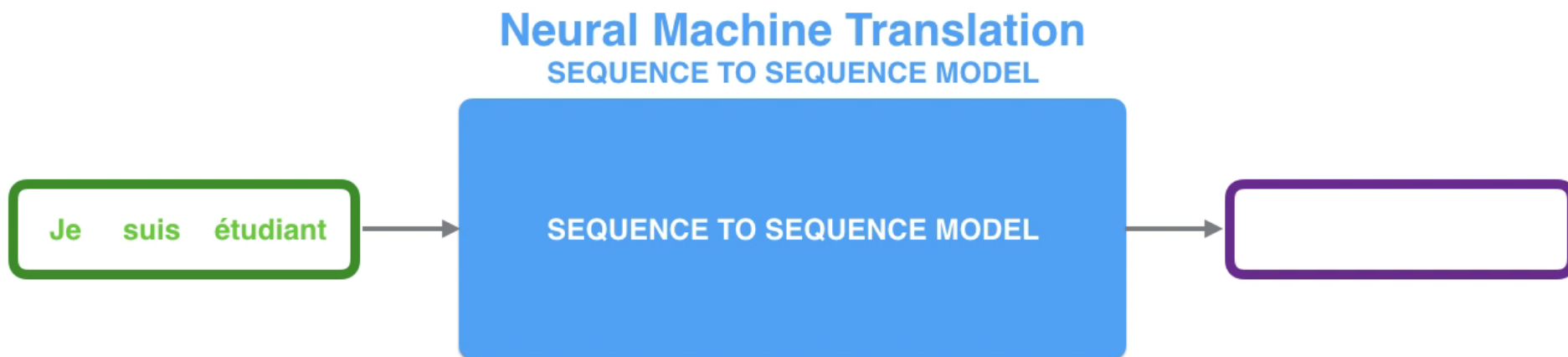


<https://jalammar.github.io/illustrated-transformer/>

<https://jalammar.github.io/visualizing-neural-machine-translation-mechanics-of-seq2seq-models-with-attention/>

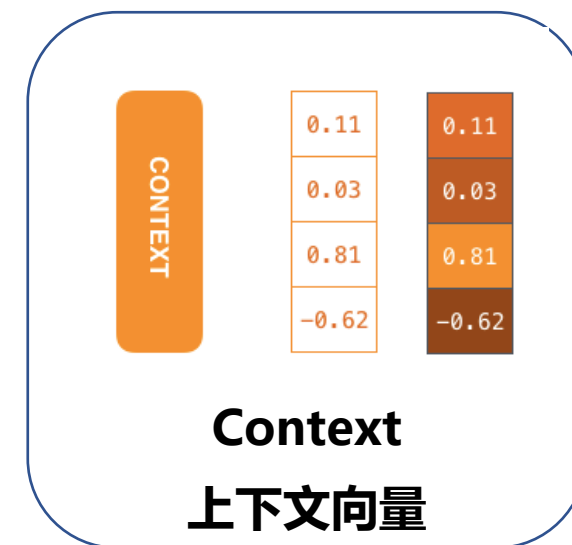
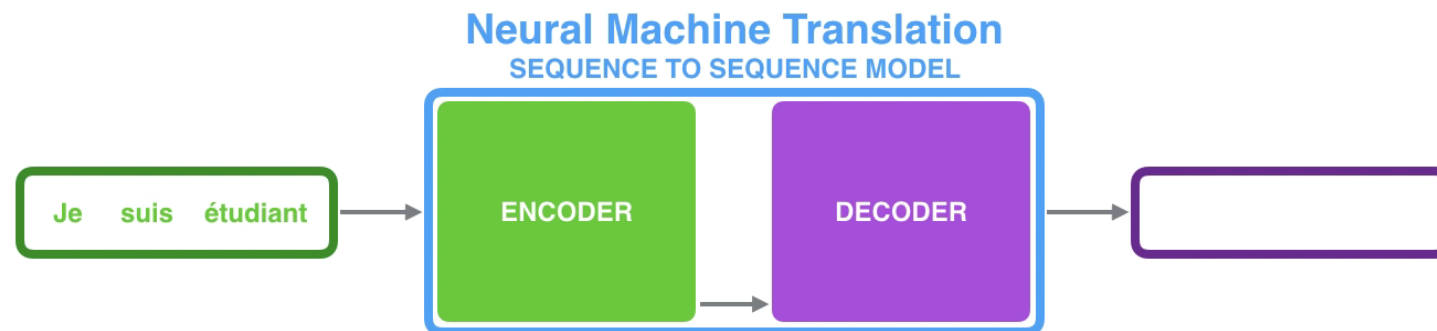
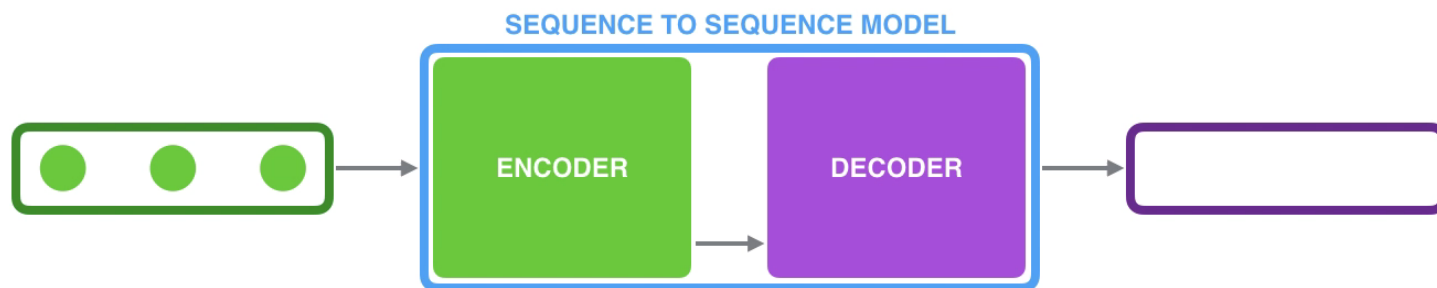
Transformer

- 序列到序列变换(seq2seq): **神经机器翻译**



Transformer

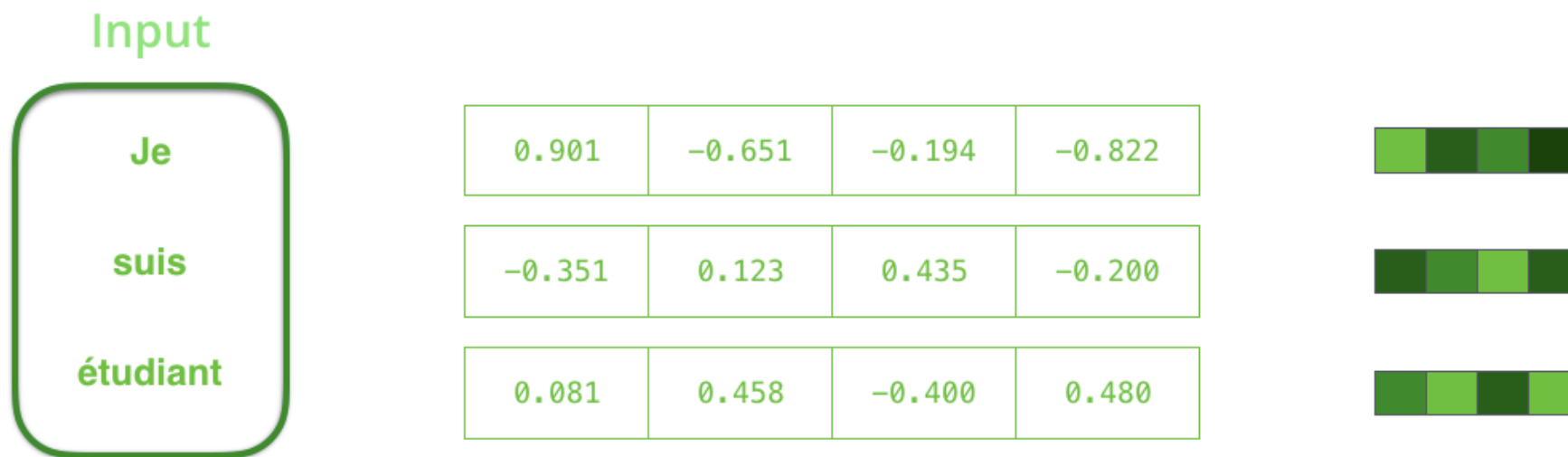
- 序列到序列变换(seq2seq): **编码解码架构**



Transformer

- 序列到序列变换(seq2seq):

- ✓ 首先将输入文本的每个单词都转为词嵌入 (如word2vec、独热)



Transformer

- 序列到序列变换(seq2seq): 回顾——基于RNN

Recurrent Neural Network

Time step #1:
An RNN takes two input vectors:

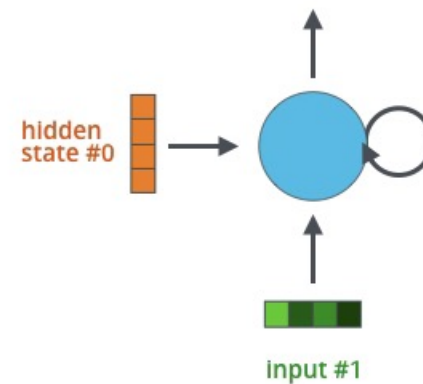


hidden state #0



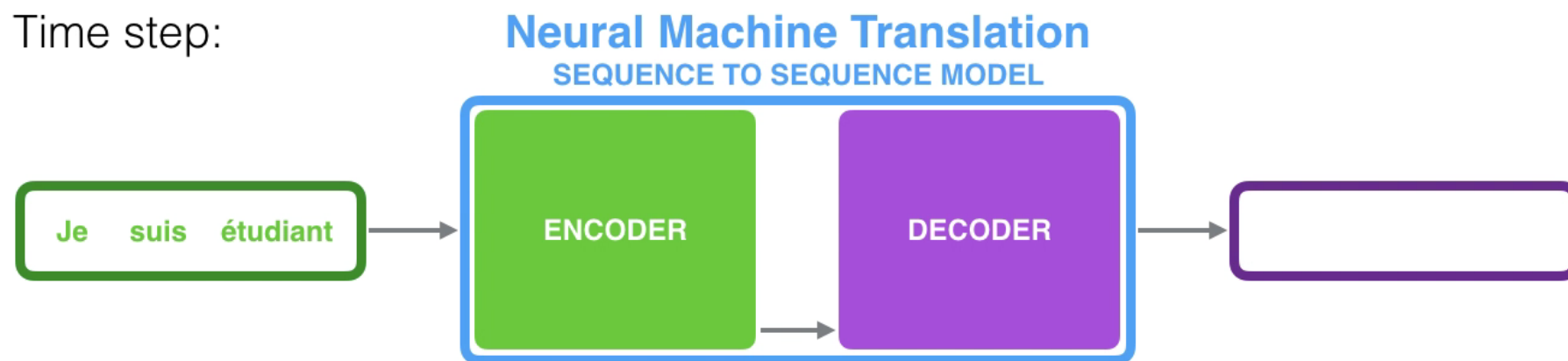
input vector #1

$$\begin{aligned} \mathbf{e}_t &= \mathbf{E}\mathbf{x}_t \\ \mathbf{h}_t &= g(\mathbf{U}\mathbf{h}_{t-1} + \mathbf{W}\mathbf{e}_t) \\ \mathbf{y}_t &= \text{softmax}(\mathbf{V}\mathbf{h}_t) \end{aligned}$$



Transformer

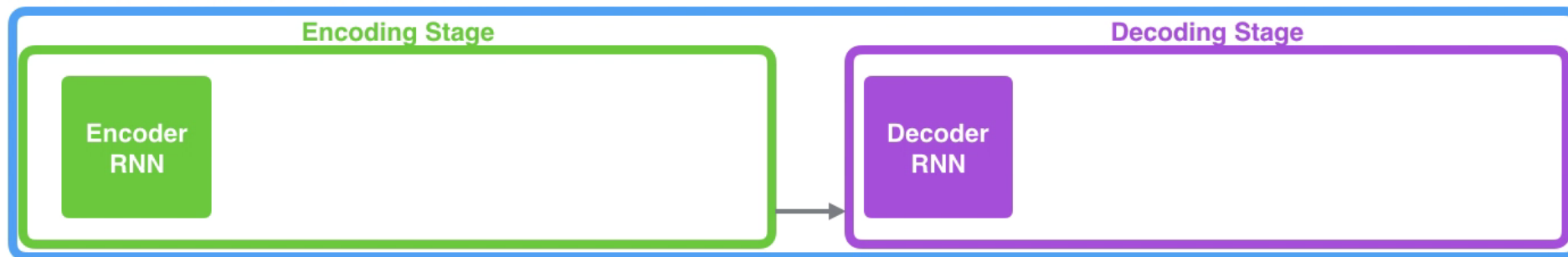
- 序列到序列变换(seq2seq): 回顾——**基于RNN**



Transformer

- 序列到序列变换(seq2seq): 回顾——基于RNN

Neural Machine Translation SEQUENCE TO SEQUENCE MODEL



Je

suis

étudiant

Transformer

- 序列到序列变换(seq2seq): **注意力机制**

Time step: 7

Neural Machine Translation

SEQUENCE TO SEQUENCE MODEL WITH ATTENTION



Transformer

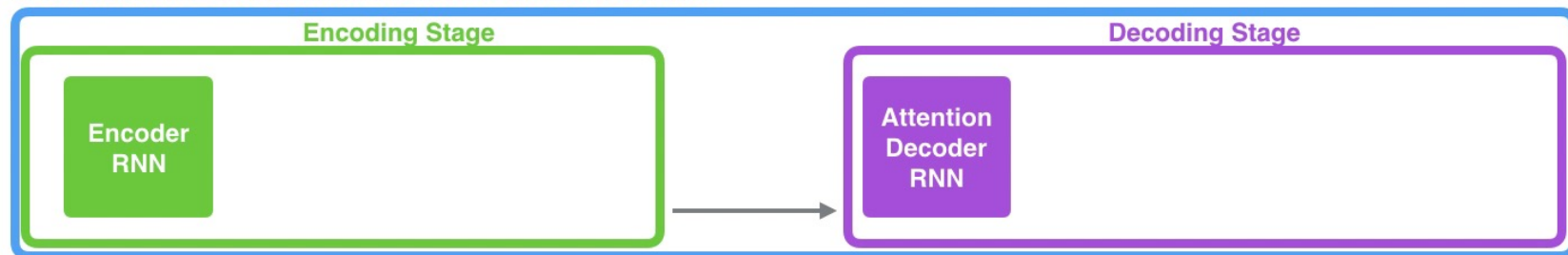
- 序列到序列变换(seq2seq): **注意力机制**

区别:

- 编码器把所有隐状态都传给解码器
- 解码器基于注意力机制对所有隐状态施加不同权重

Neural Machine Translation

SEQUENCE TO SEQUENCE MODEL WITH ATTENTION



Je

suis

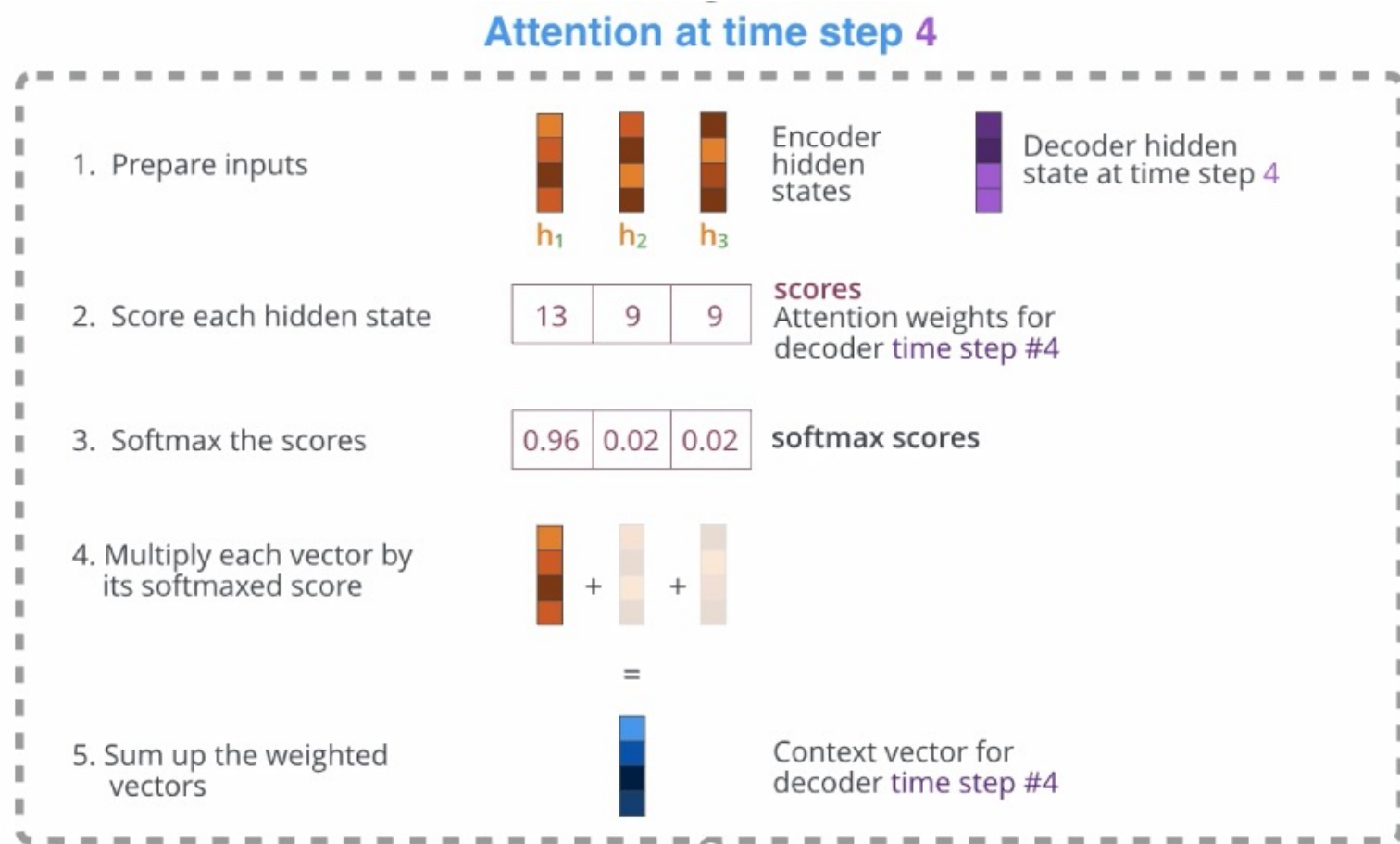
étudiant

Transformer

- 序列到序列变换(seq2seq): **注意力机制**

区别:

- 编码器把所有隐状态都传给解码器
- 解码器基于注意力机制对所有隐状态进行加权求和



Transformer

- 序列到序列变换(seq2seq): **注意力机制**

区别:

- 编码器把所有隐状态都传给解码器
- 解码器基于注意力机制对所有隐状态进行加权求和

Attention at time step 4

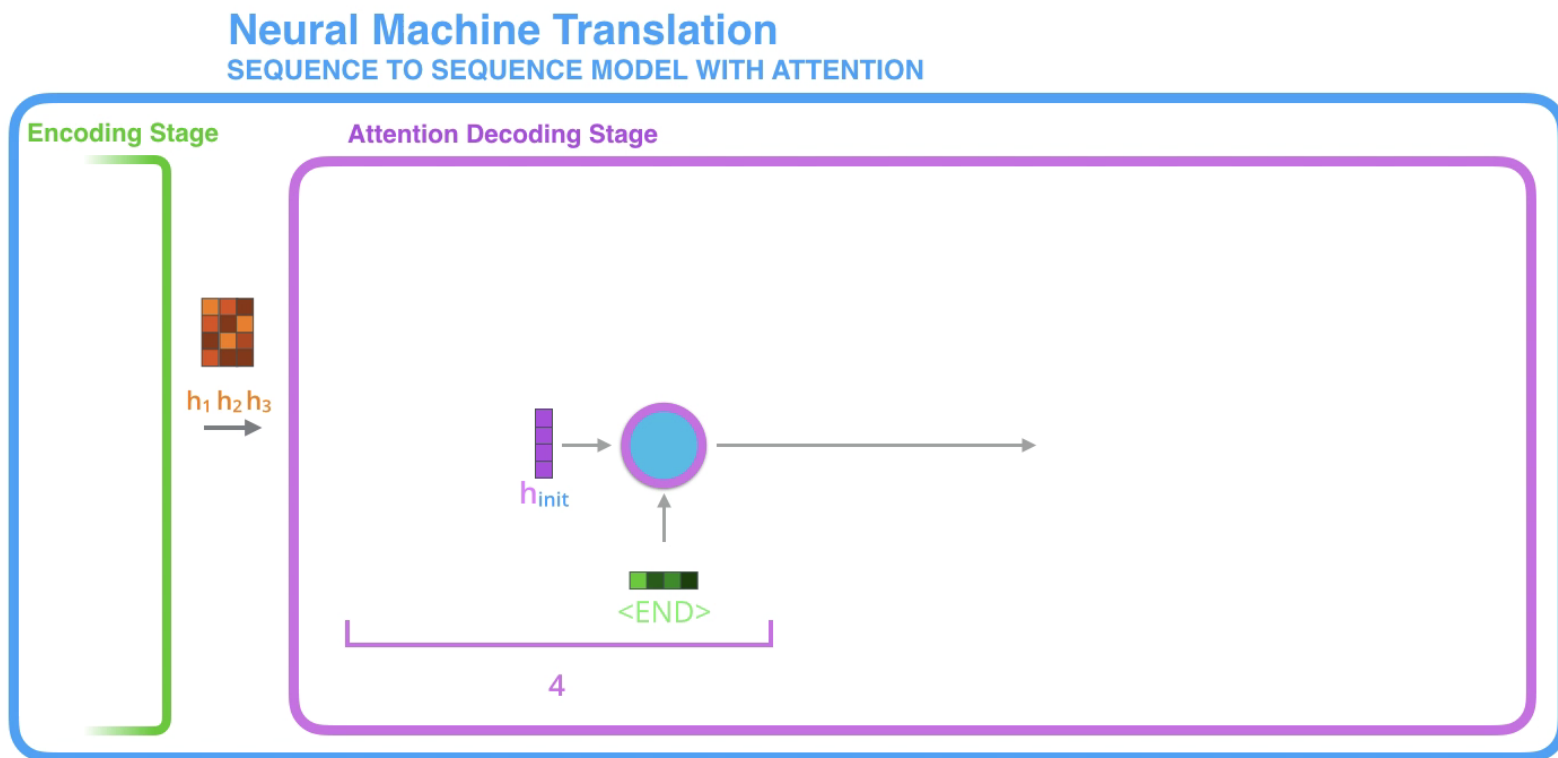


Transformer

- 序列到序列变换(seq2seq): **注意力机制**

区别:

- 编码器把所有隐状态都传给解码器
- 解码器基于注意力机制对所有隐状态进行加权求和

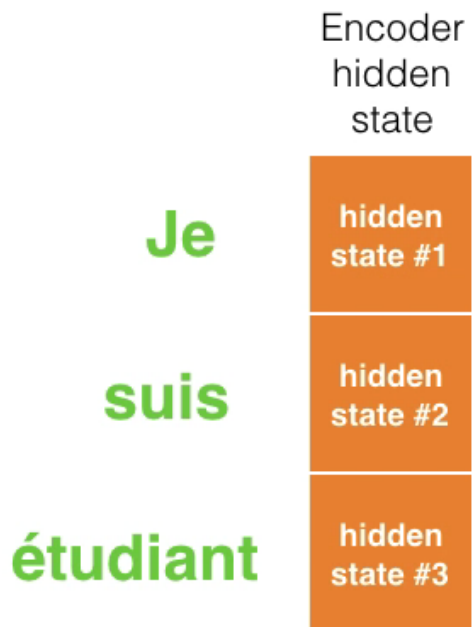


Transformer

- 序列到序列变换(seq2seq): **注意力机制**

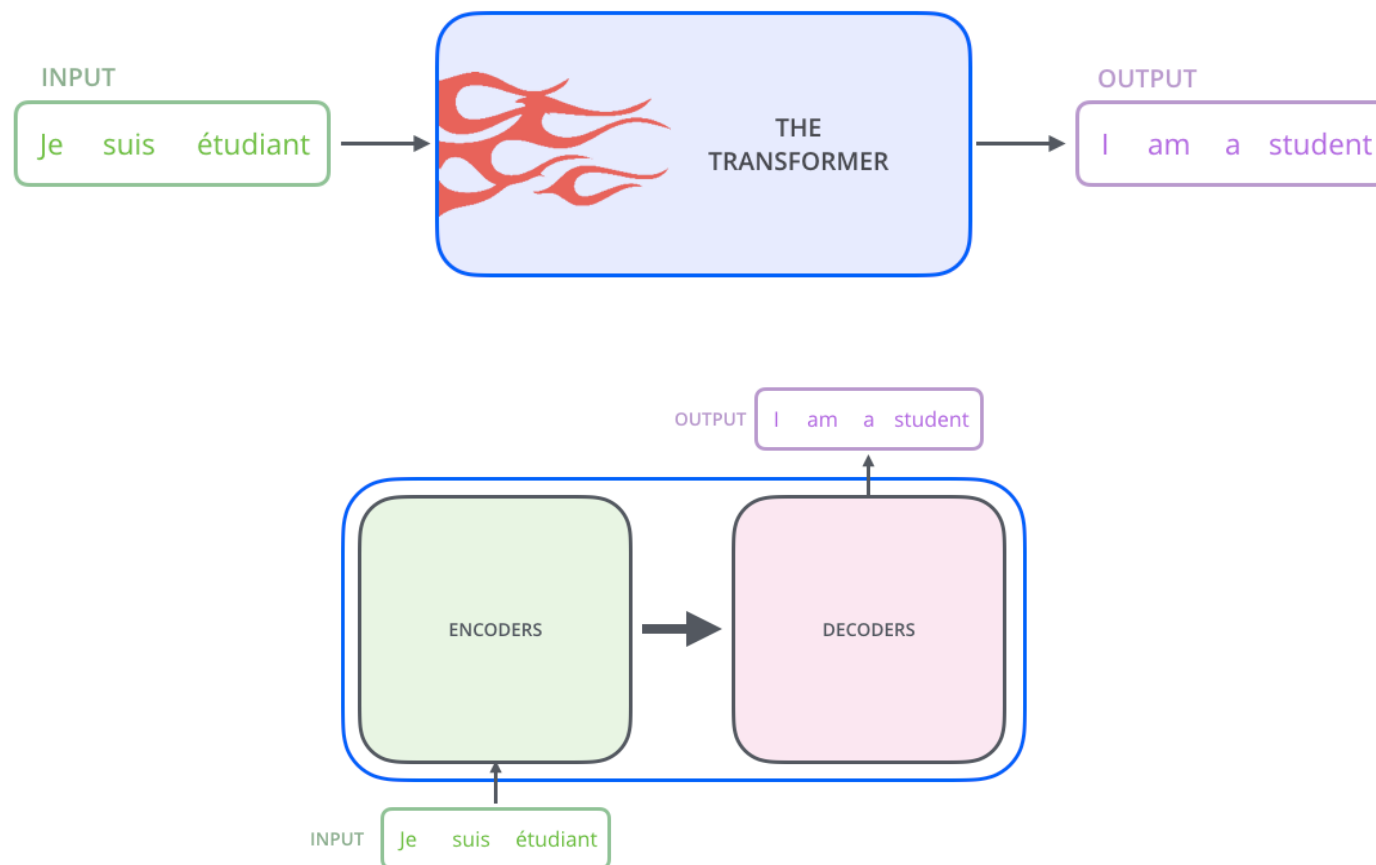
区别:

- 编码器把所有隐状态都传给解码器
- 解码器基于注意力机制对所有隐状态进行加权求和



Transformer

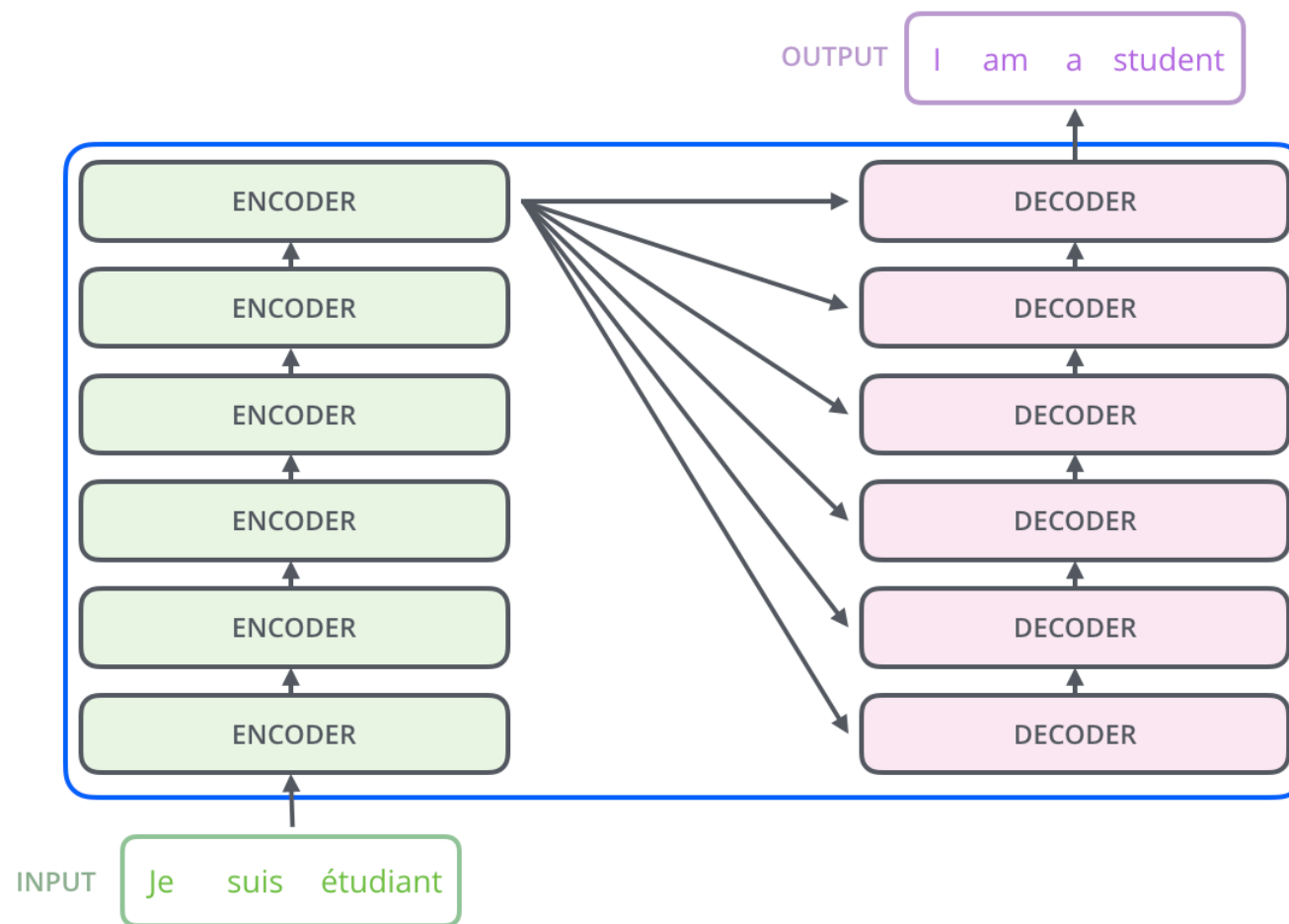
- **注意力机制**: Attention is all you need



Attention is all you need, Vaswani et al. NeurIPS 2017

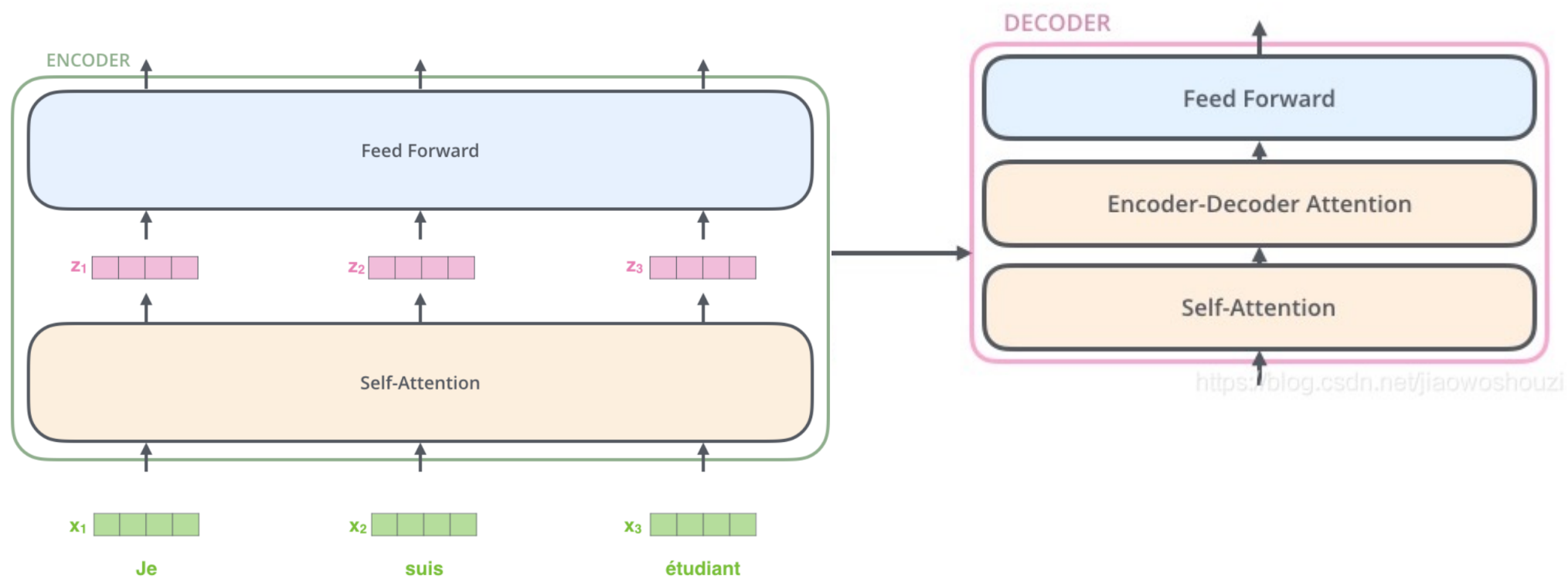
Transformer

- **注意力机制**: Attention is all you need



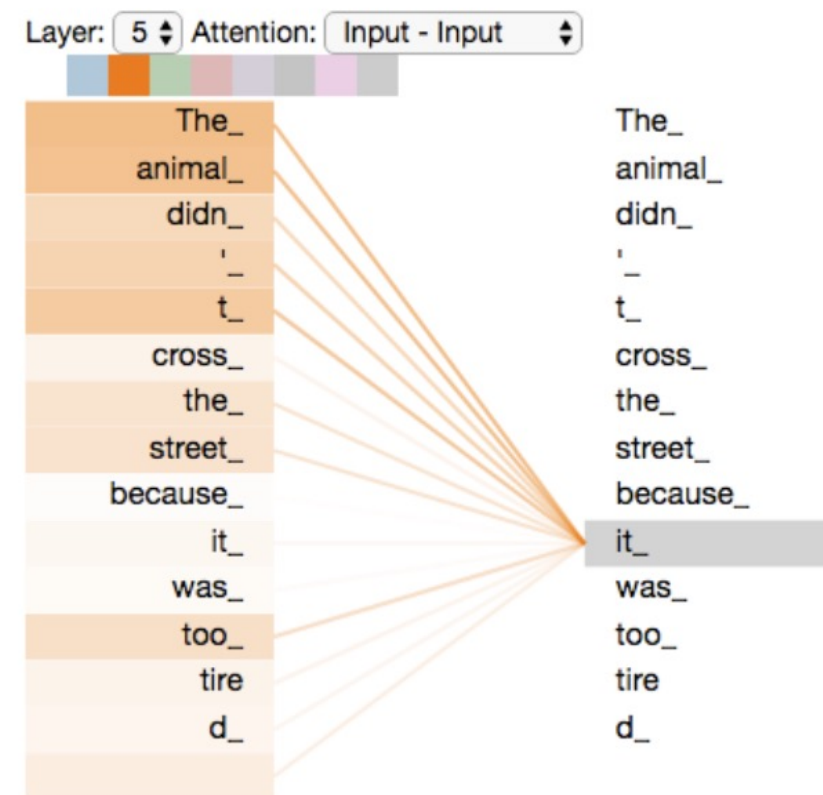
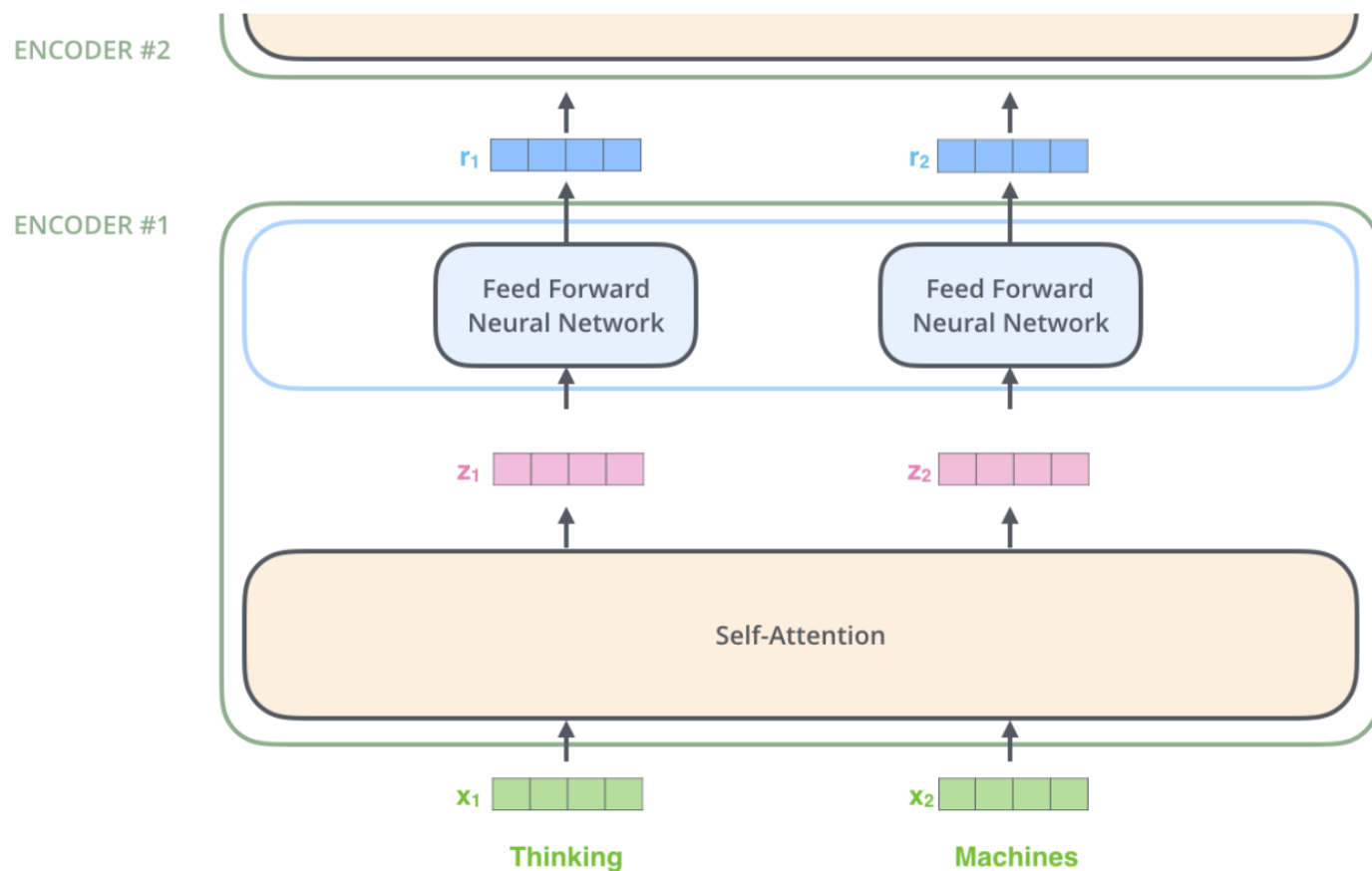
Transformer

- **注意力机制**: Attention is all you need



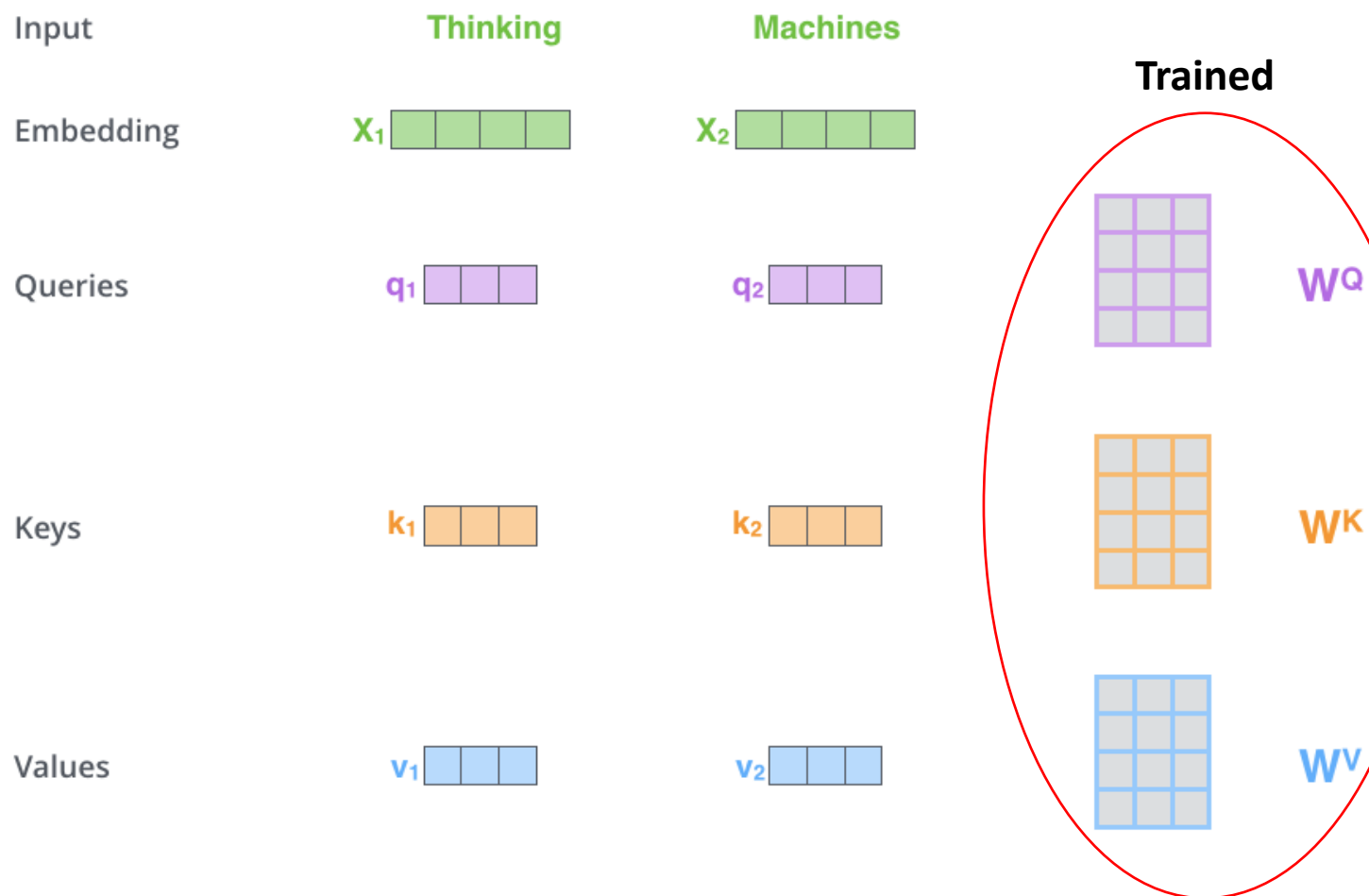
Transformer

- **注意力机制**: Attention is all you need



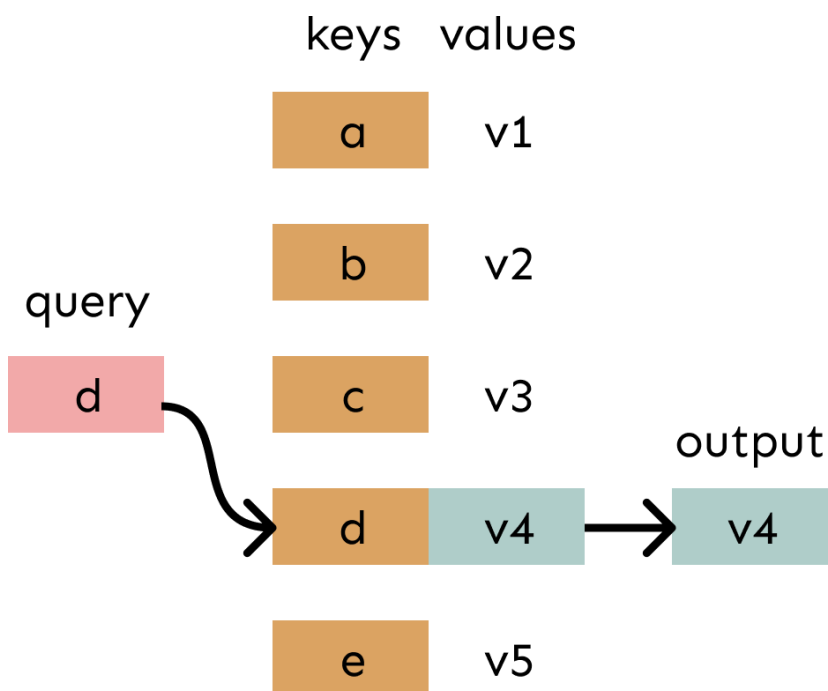
Transformer

- **注意力机制**: Attention is all you need

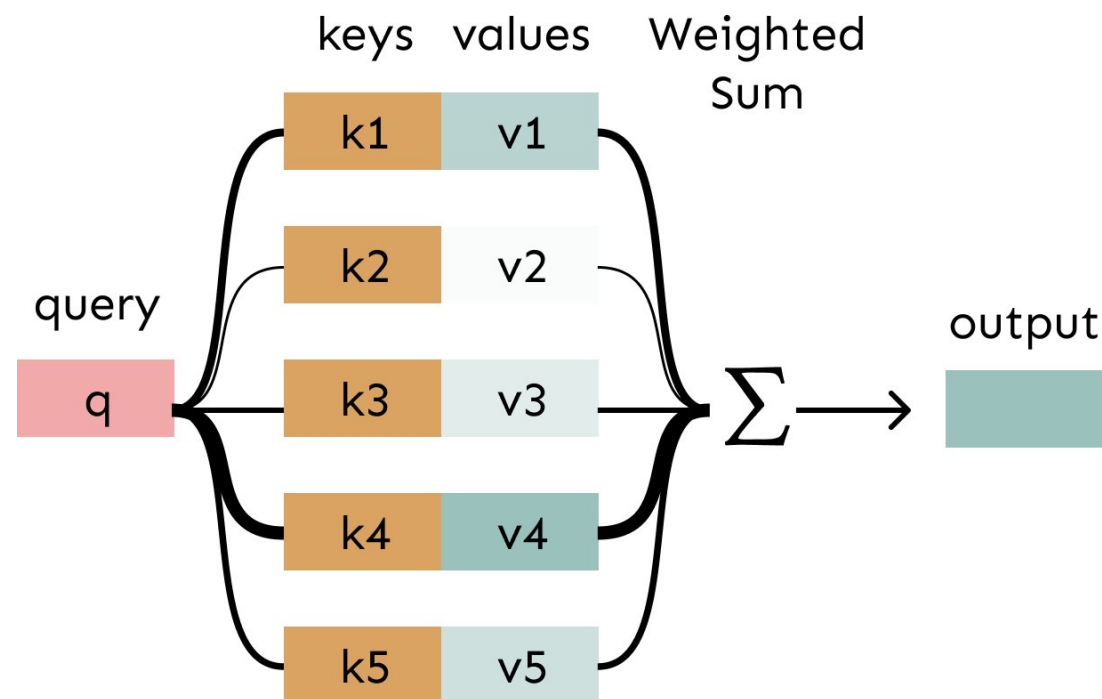


Transformer——注意力机制(attention)

我们可以将注意力机制当成模糊查找表。在这个查找表中**keys**和**values**——匹配。当**query**匹配上其中的某个keys,就返回对应的value。



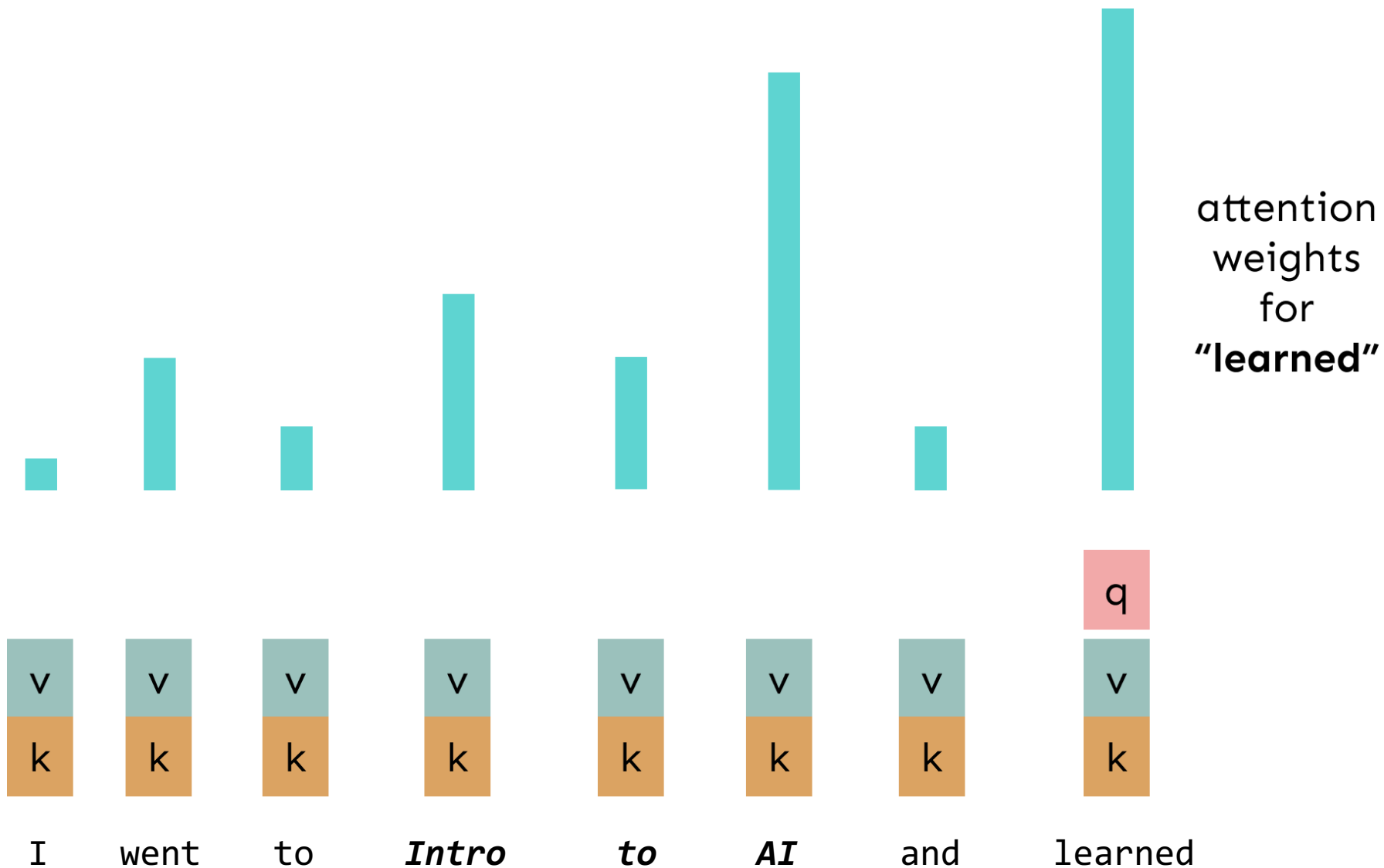
在注意力机制中，**query**会与所有的**keys**匹配，权重介于0和1之间。输出的**values**是所有value加权平均。



Transformer——注意力机制(attention)



北京大学
PEKING UNIVERSITY



Transformer——注意力机制(attention)

$w_{1:n}$ 是一个由词汇表 V 组成的句子，例如 *Zuko made his uncle tea*.

对于每个 w_i ，求得其词嵌入表示向量 $x_i = e(w_i)$

1. 给每个词嵌入表示乘以维度为 $\mathbb{R}^{d \times d}$ 的权重矩阵 W^Q, W^K, W^V

$$q_i = x_i W^Q (\text{queries})$$

$$k_i = x_i W^K (\text{keys})$$

$$v_i = x_i W^V (\text{values})$$

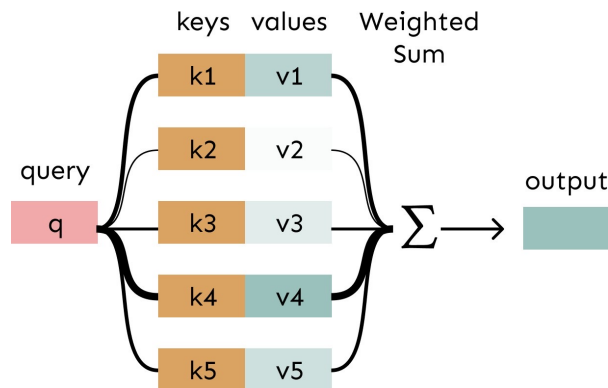
2. 计算每一对query和key的相似度，然后进行normalize 和 softmax

$$e_{ij} = q_i^T k_j$$

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_j \exp(e_{ij})}$$

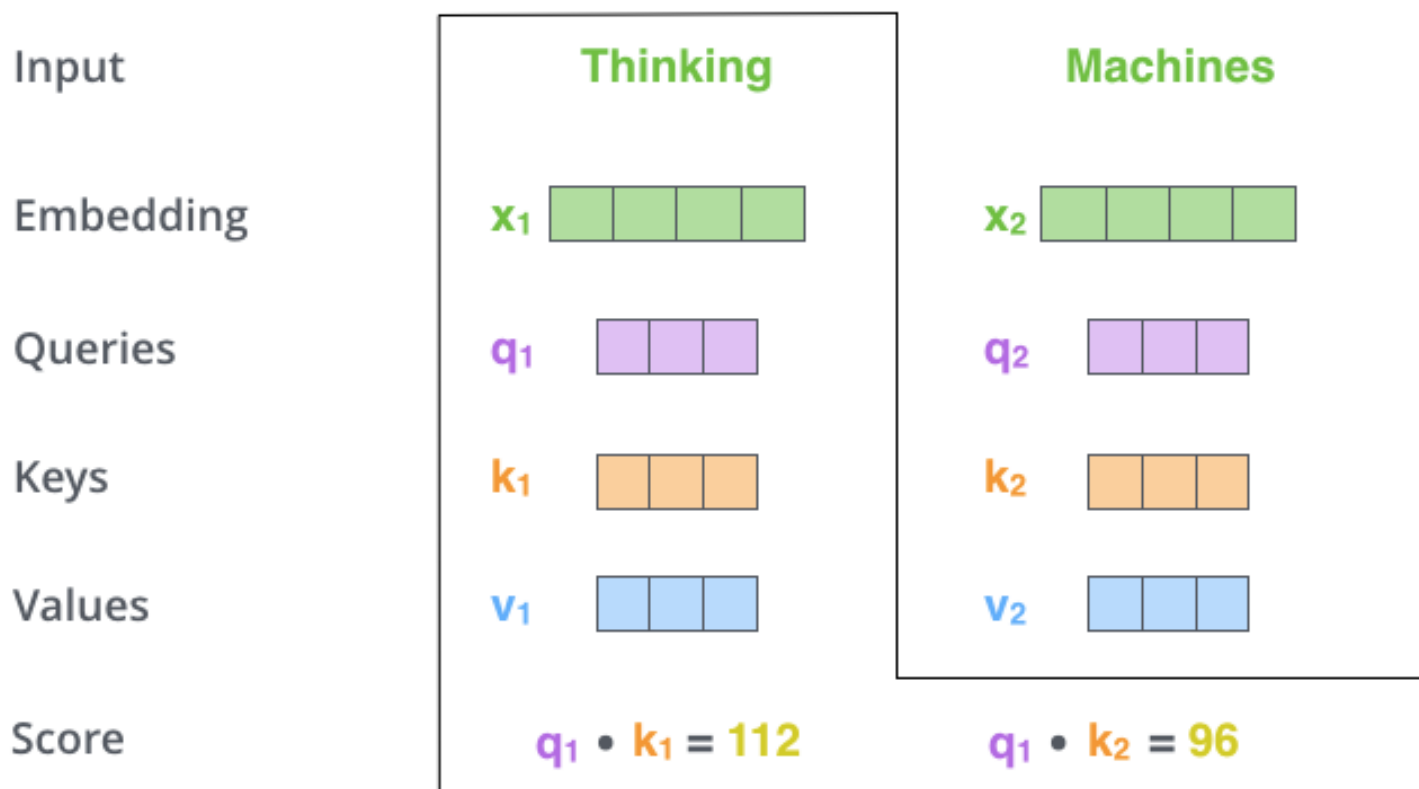
3. 计算values的加权平均作为每个词的输出

$$o_i = \sum_j \alpha_{ij} v_j$$



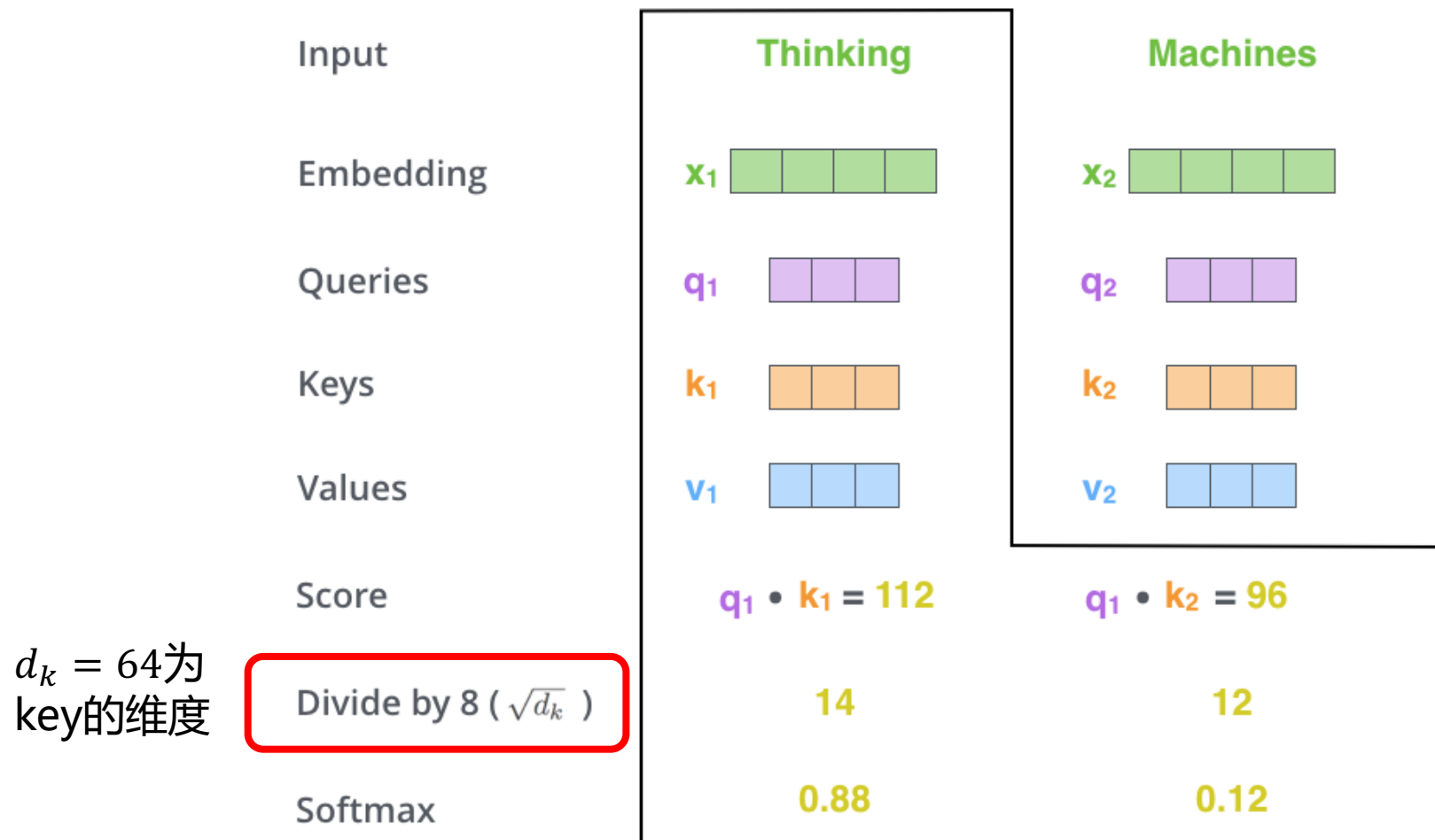
Transformer

- **注意力机制**: Attention is all you need



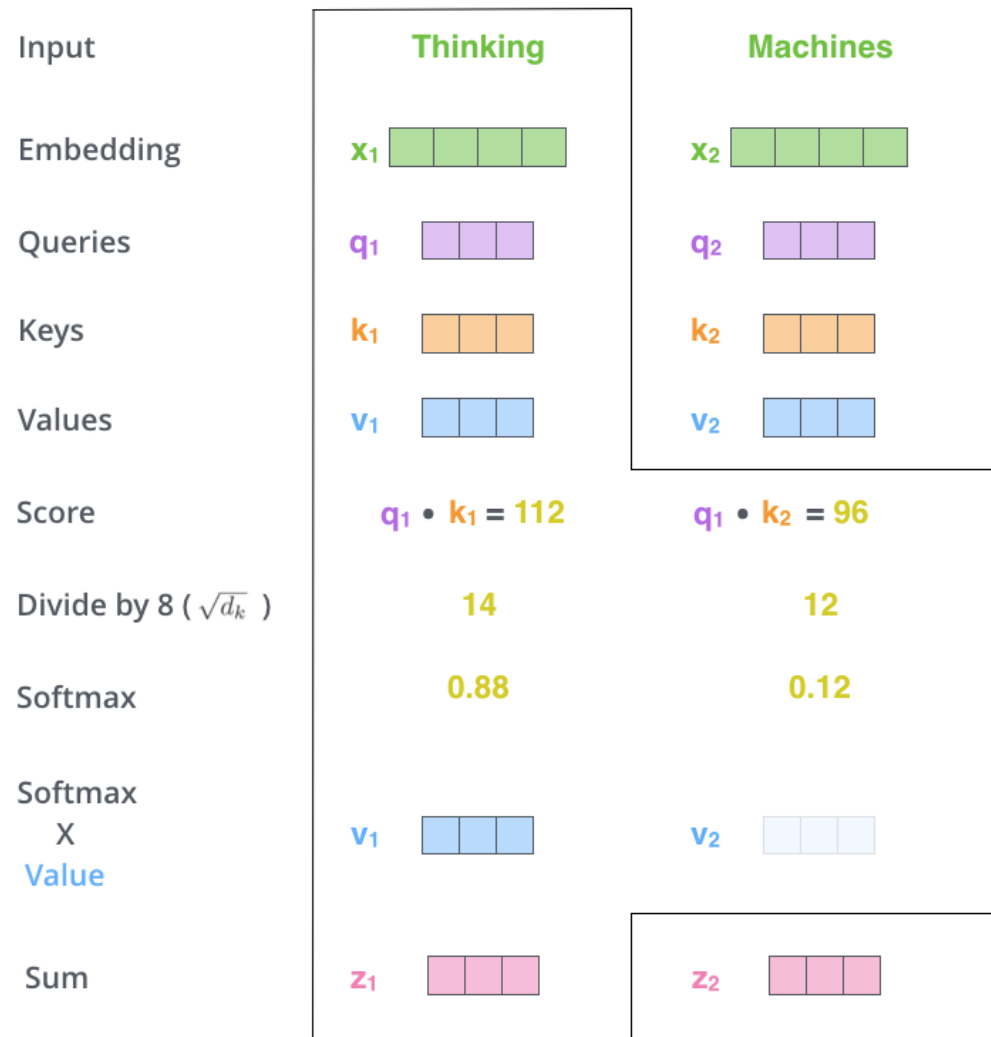
Transformer

- **注意力机制**: Attention is all you need



Transformer

- **注意力机制**: Attention is all you need



Transformer

• 注意力机制：矩阵运算

$$\begin{matrix} X \\ \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline \end{array} \end{matrix} \times \begin{matrix} W^Q \\ \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline \end{array} \end{matrix} = \begin{matrix} Q \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array} \end{matrix}$$

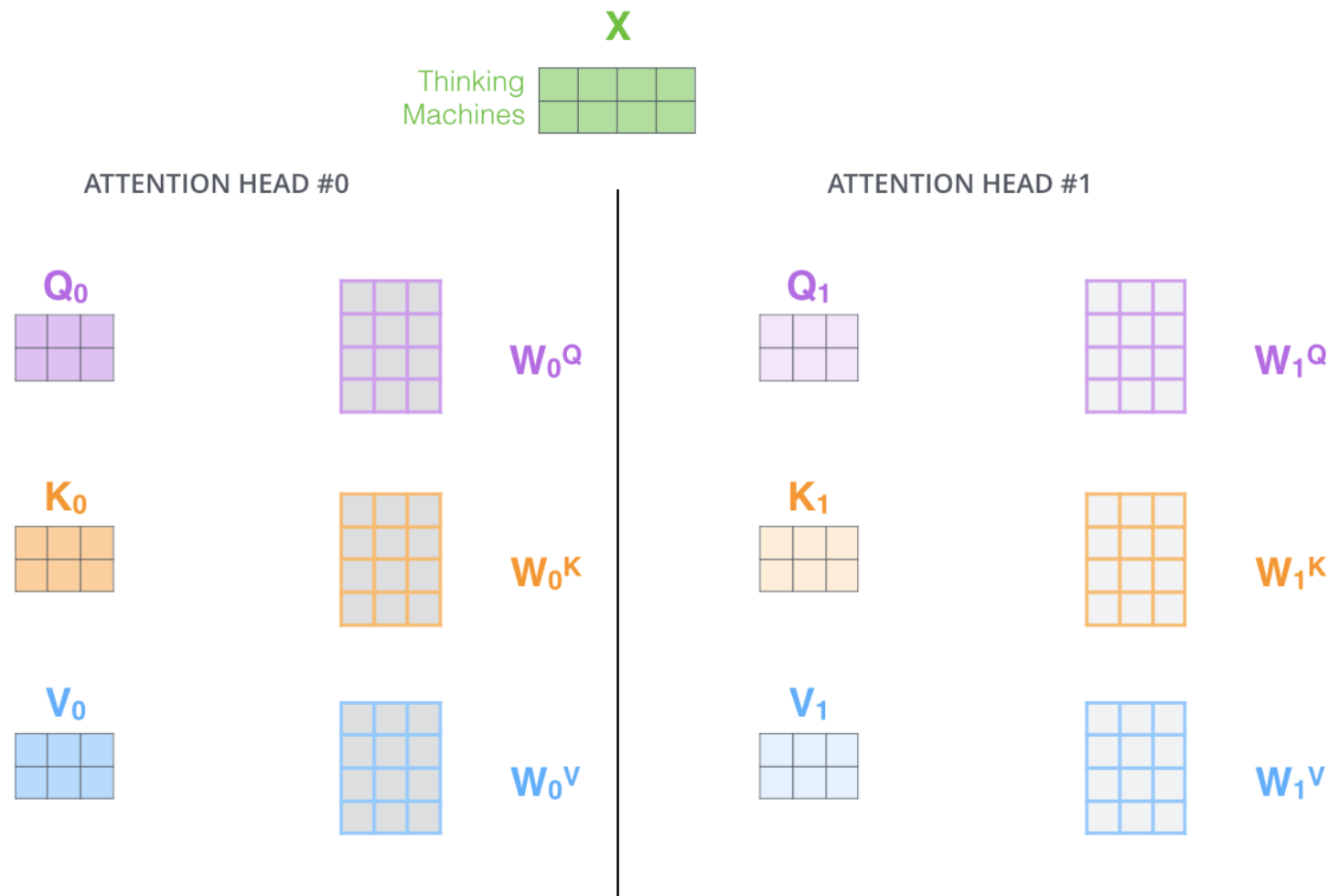
$$\begin{matrix} X \\ \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline \end{array} \end{matrix} \times \begin{matrix} W^K \\ \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline \end{array} \end{matrix} = \begin{matrix} K \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array} \end{matrix}$$

$$\begin{matrix} X \\ \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline \end{array} \end{matrix} \times \begin{matrix} W^V \\ \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline \end{array} \end{matrix} = \begin{matrix} V \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array} \end{matrix}$$

$$\text{softmax} \left(\frac{\begin{matrix} Q \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array} \end{matrix} \times \begin{matrix} K^T \\ \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \end{matrix}}{\sqrt{d_k}} \right) \begin{matrix} V \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array} \end{matrix}$$
$$= \begin{matrix} Z \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array} \end{matrix}$$

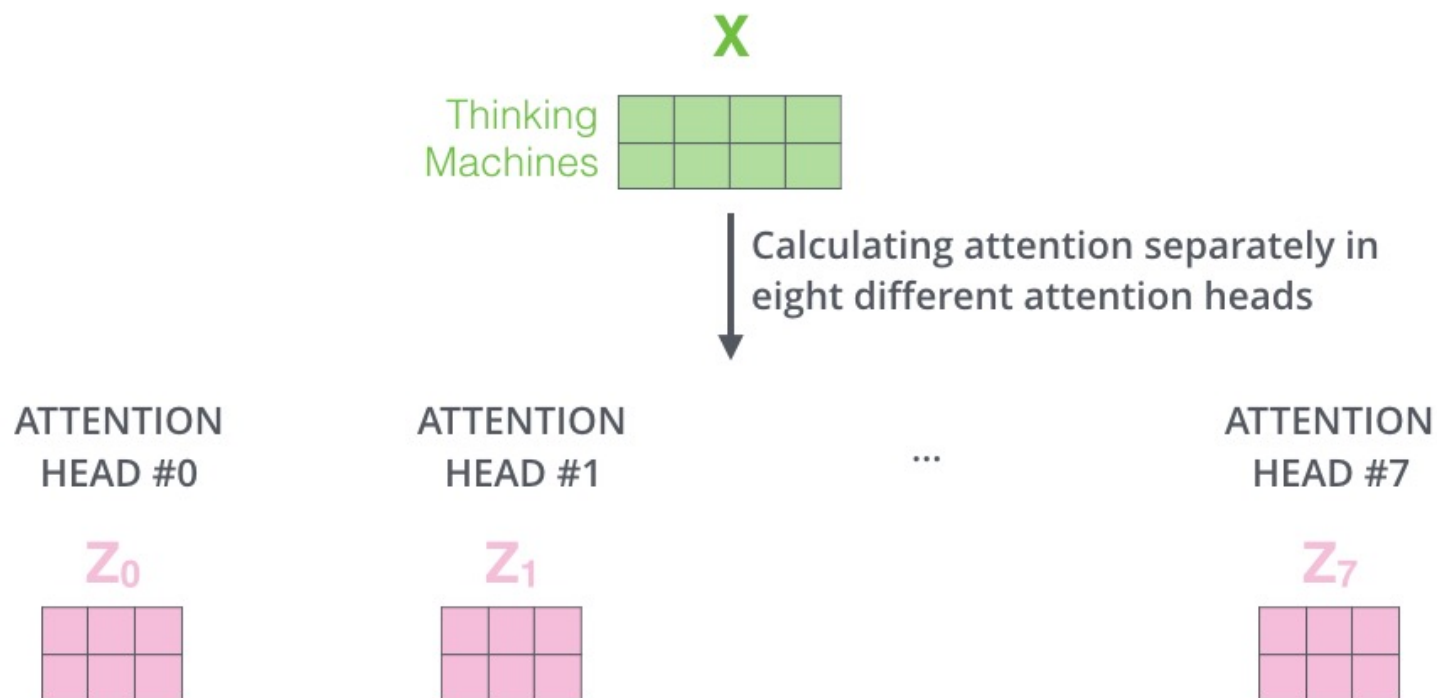
Transformer

- **注意力机制**：矩阵运算+多头注意力



Transformer

- **注意力机制**：矩阵运算+多头注意力

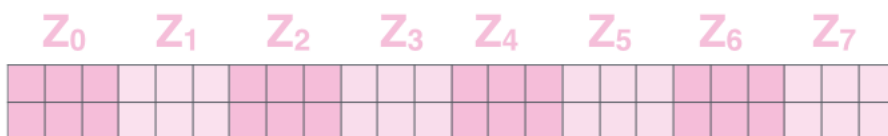


Transformer



• 注意力机制：矩阵运算+多头注意力

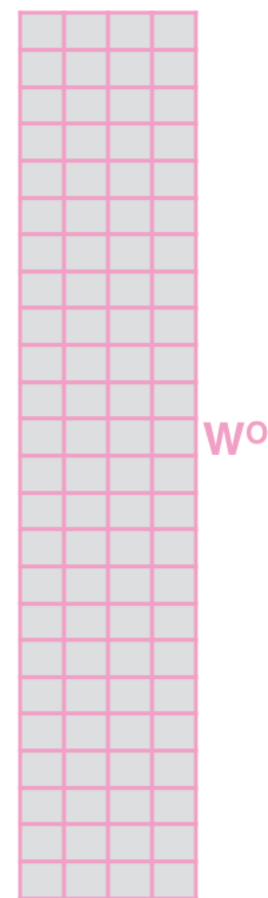
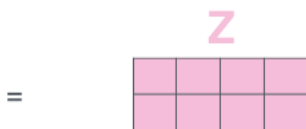
1) Concatenate all the attention heads



2) Multiply with a weight matrix W^O that was trained jointly with the model

x

3) The result would be the Z matrix that captures information from all the attention heads. We can send this forward to the FFNN



Transformer



• 注意力机制：矩阵运算+多头注意力

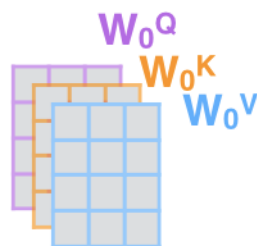
1) This is our
input sentence*

Thinking
Machines

2) We embed
each word*



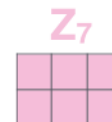
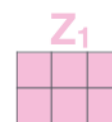
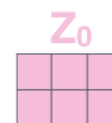
3) Split into 8 heads.
We multiply X or
 R with weight matrices



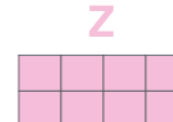
4) Calculate attention
using the resulting
 $Q/K/V$ matrices



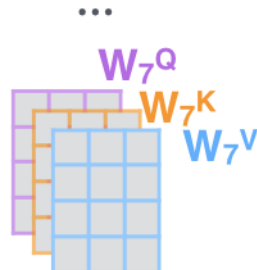
5) Concatenate the resulting Z matrices,
then multiply with weight matrix W^O to
produce the output of the layer



W^O

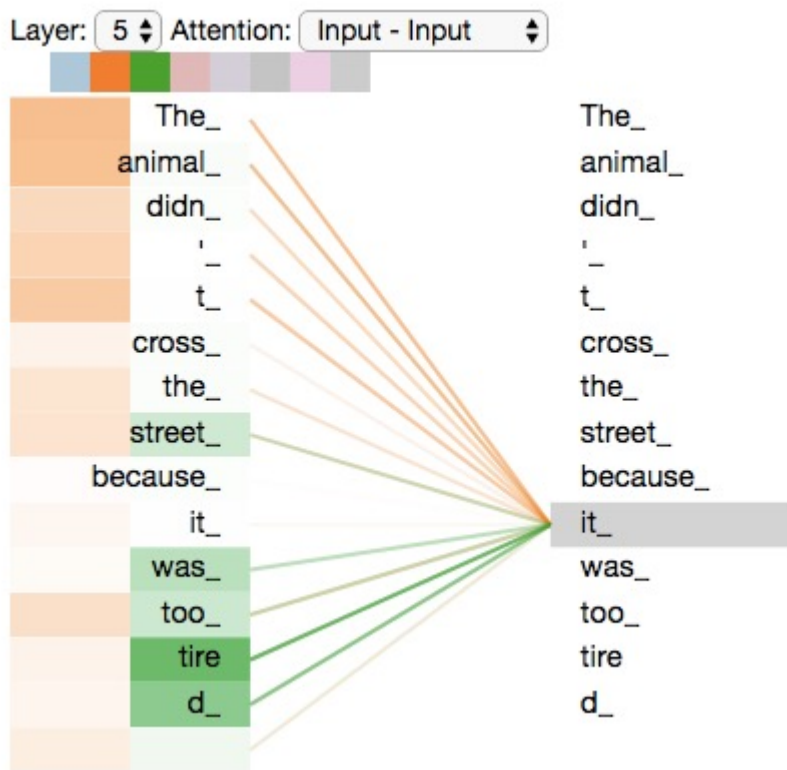


* In all encoders other than #0,
we don't need embedding.
We start directly with the output
of the encoder right below this one

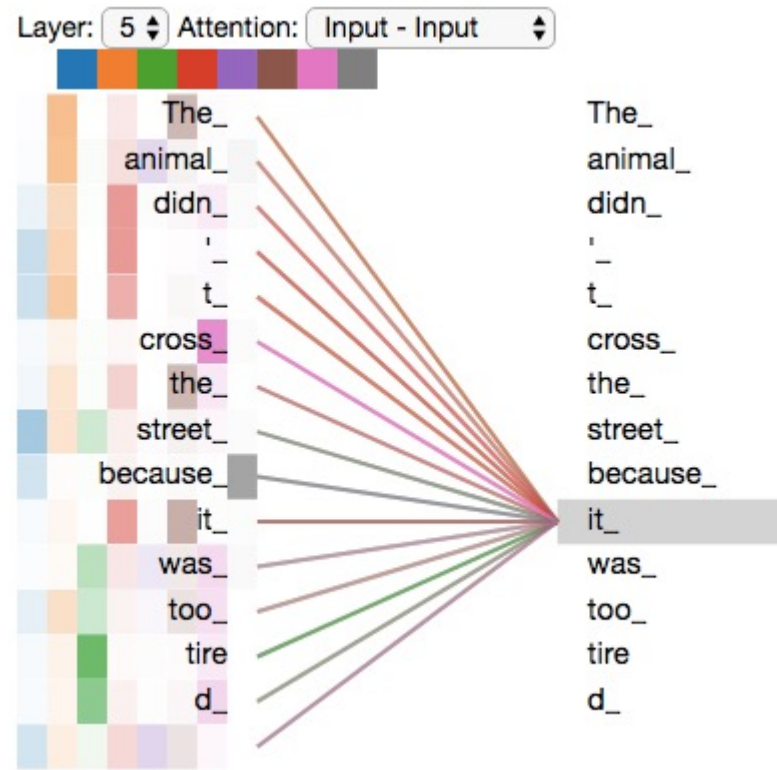


Transformer

- 注意力机制：多头注意力



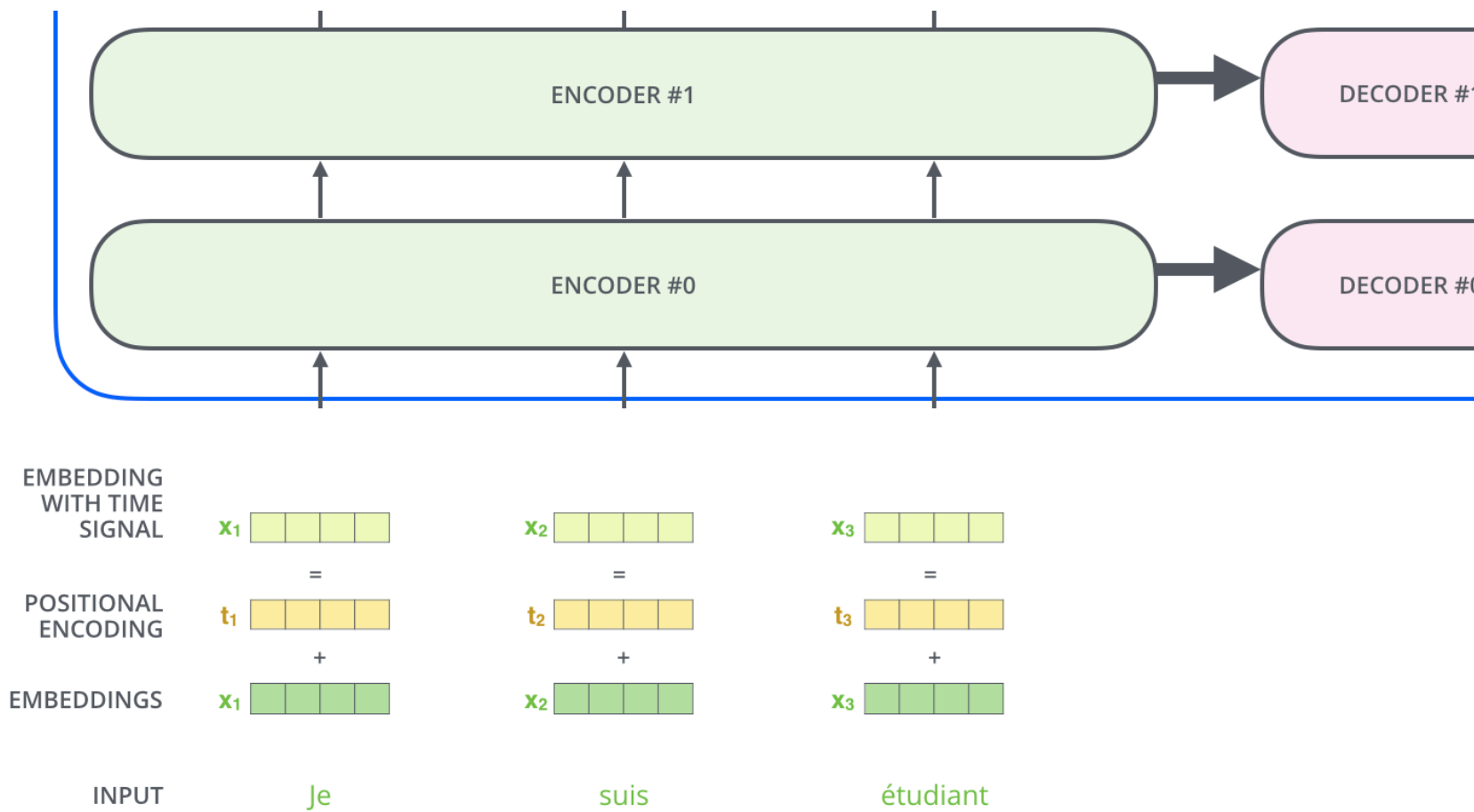
Two attention heads



All attention heads

Transformer

• 注意力机制：位置编码

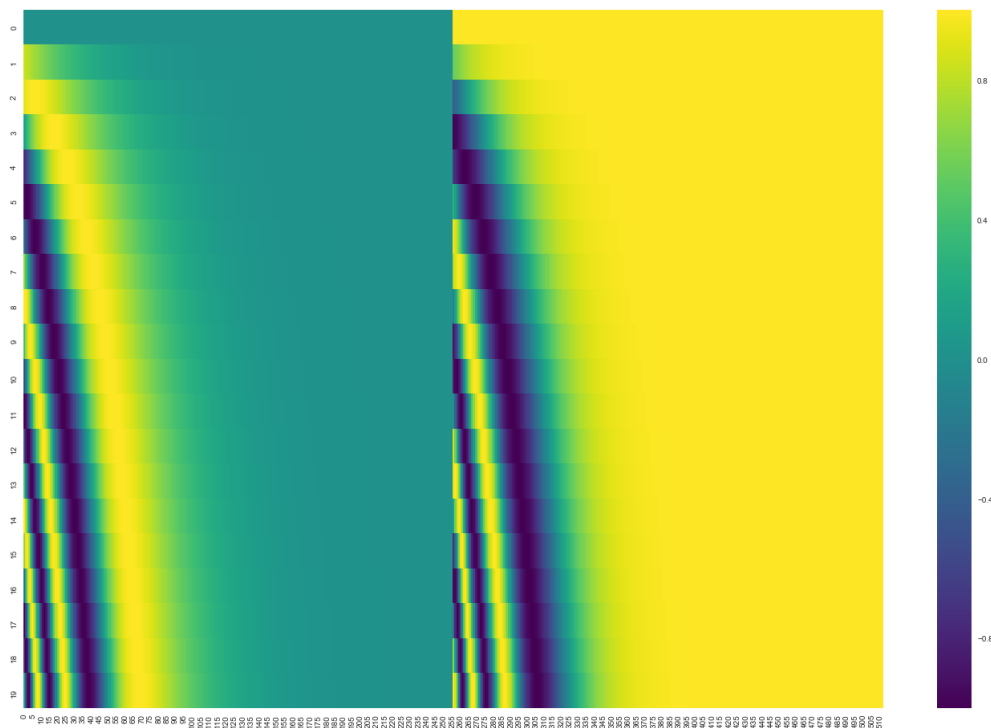


Transformer

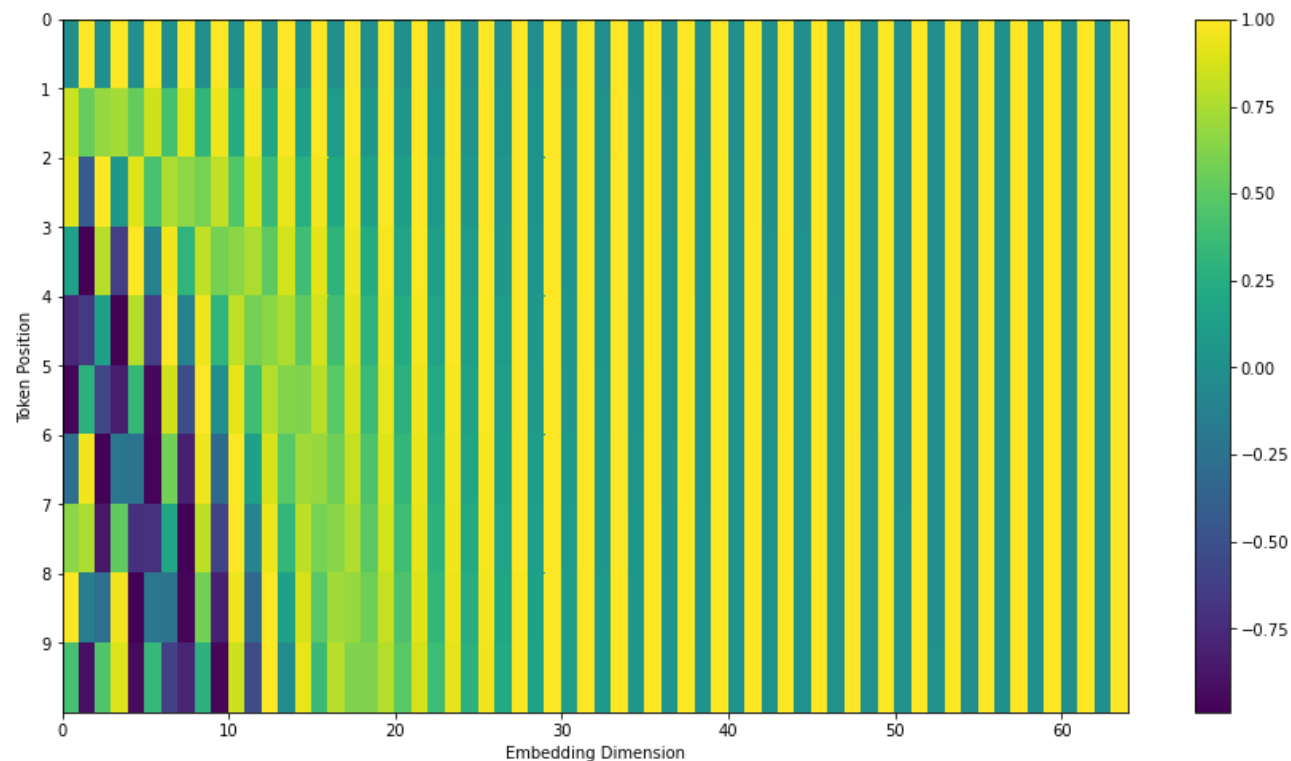
• 注意力机制：位置编码

$$\text{PE}(\text{pos}, 2i) = \sin\left(\frac{\text{pos}}{10000^{2i/d}}\right)$$

$$\text{PE}(\text{pos}, 2i + 1) = \cos\left(\frac{\text{pos}}{10000^{2i/d}}\right)$$



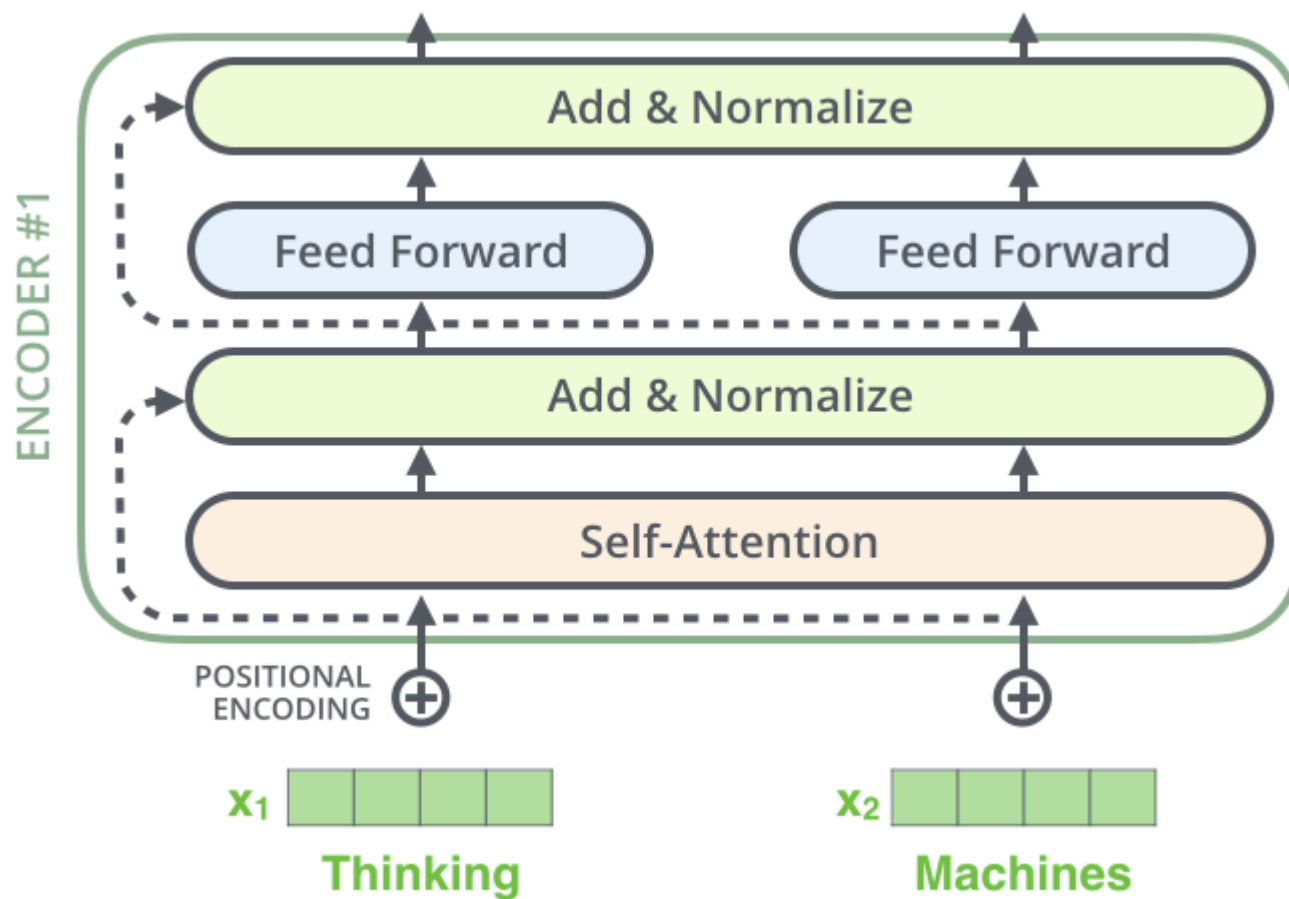
512维向量



64维向量

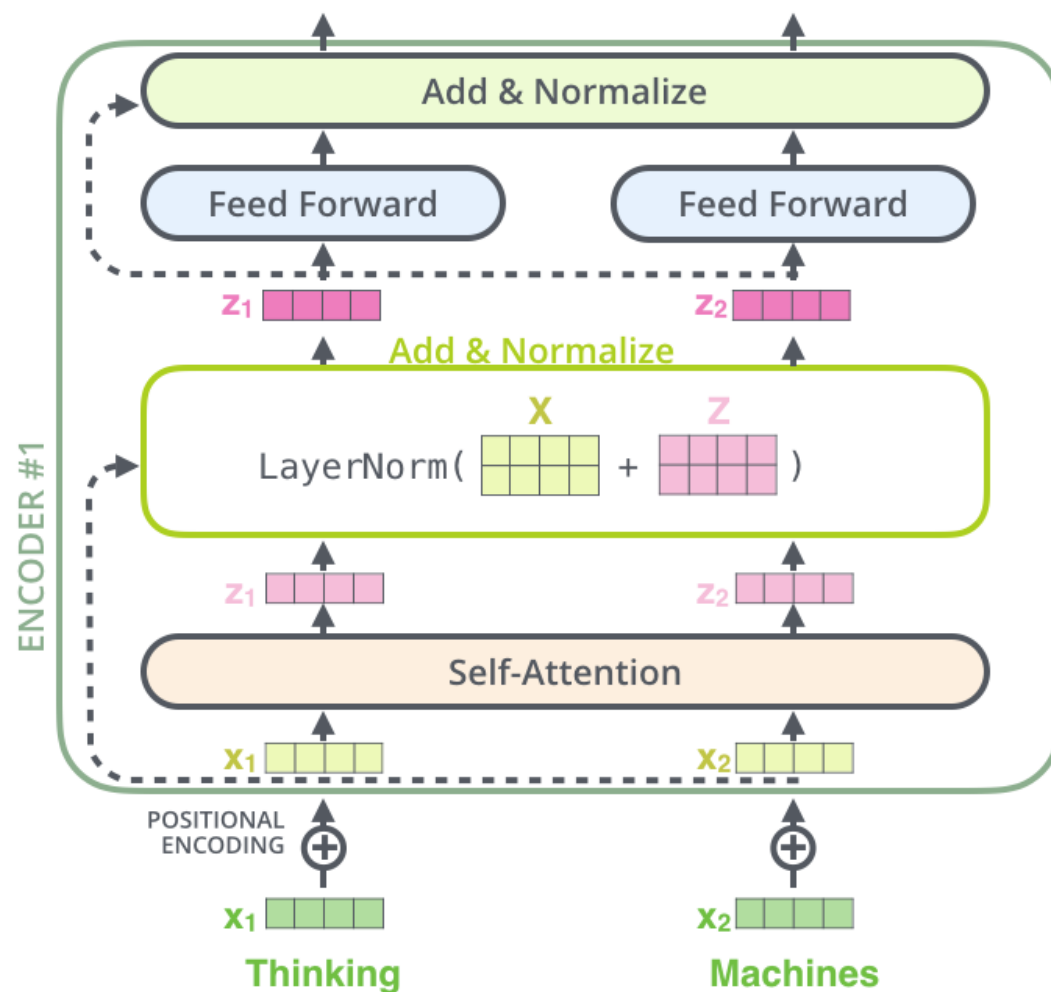
Transformer

- **注意力机制**：残差连接与layer normalization



Transformer

- **注意力机制**：残差连接与layer normalization



Transformer——一些技巧的详细描述

- 当嵌入表示的维度 d 变得很大时，点乘的结果会变得很大。Softmax的输入也会很大，导致梯度很小。
- 因此给注意力结果上乘一个 $\sqrt{d}/h$ (**Scaled Dot Product**)，则输出变为：

$$output = softmax\left(\frac{XW^Q(W^K)^T X^T}{\sqrt{d/h}}\right) \times XW^V$$

h表示head数量

- 和ResNet一样，利用**残差连接**能够帮助Transformer更好的训练。
- 类似于CNN中使用批归一化，Transformer使用**层归一化**帮助训练：

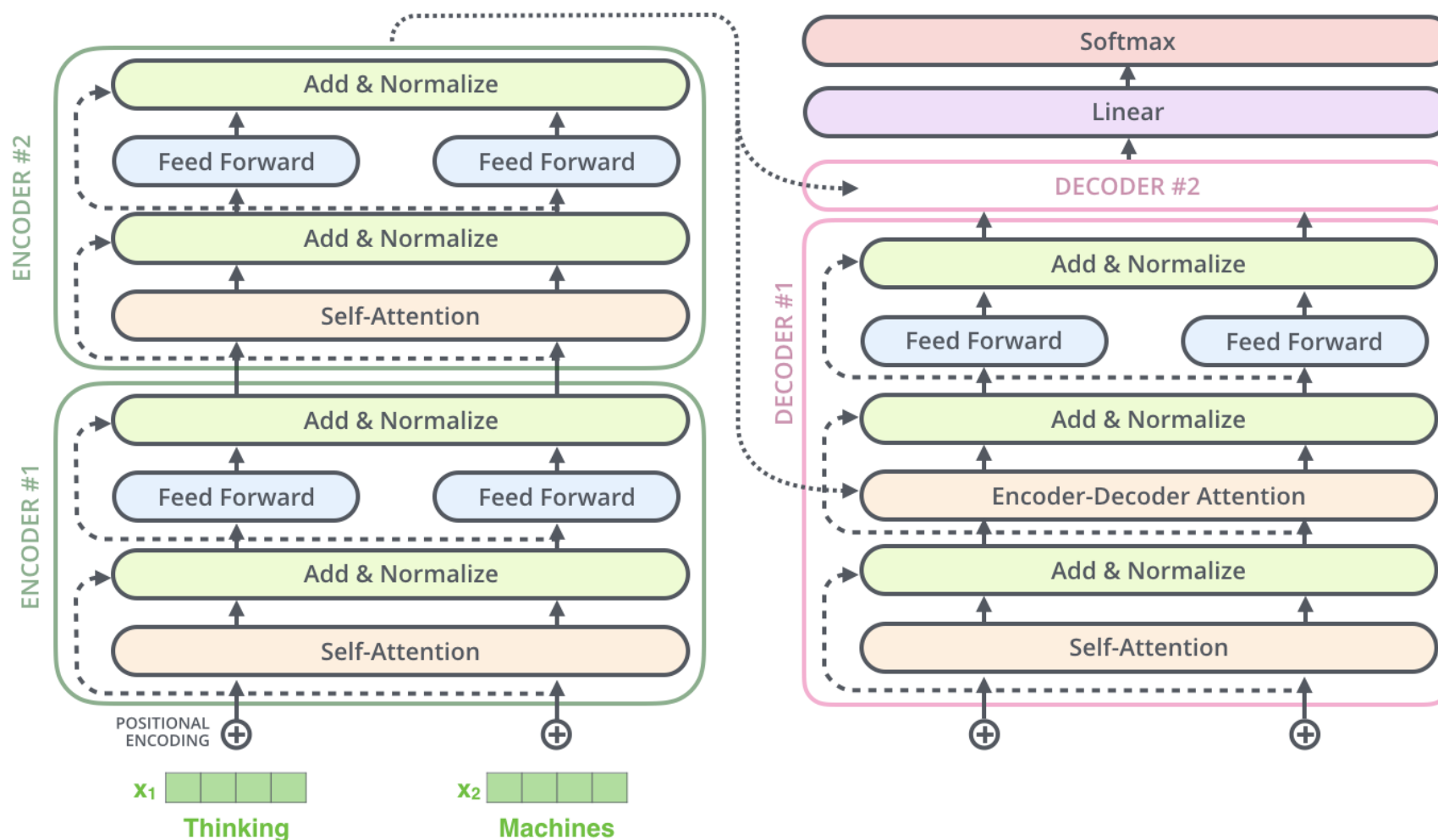
$$output = \frac{x - \mu}{\sqrt{\sigma + \epsilon}} * \gamma + \beta$$

Normalize by scalar mean and variance

Modulate by learned elementwise gain and bias

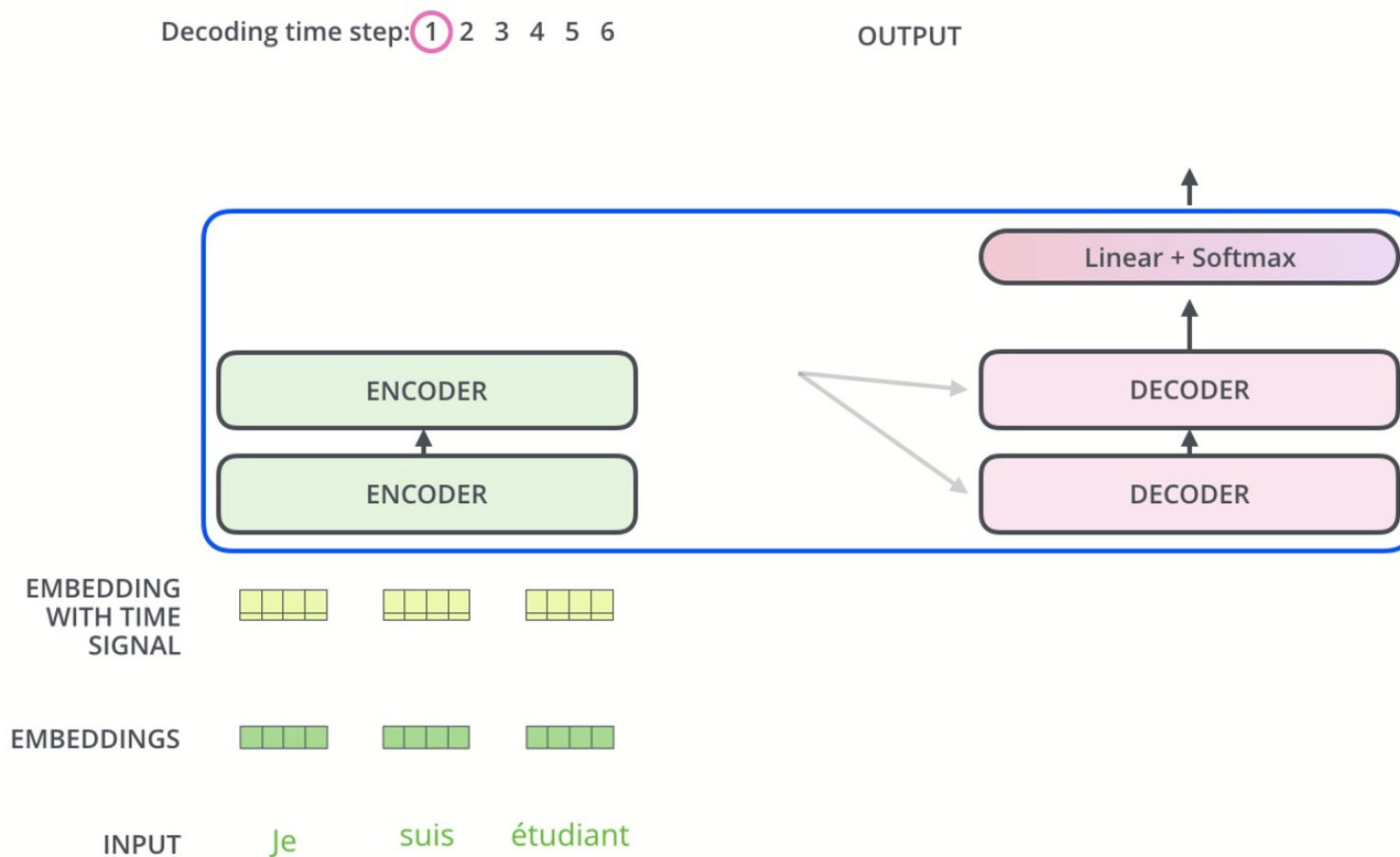
Transformer

- 由2个堆叠编码器与2个堆叠解码器组成的Transformer:



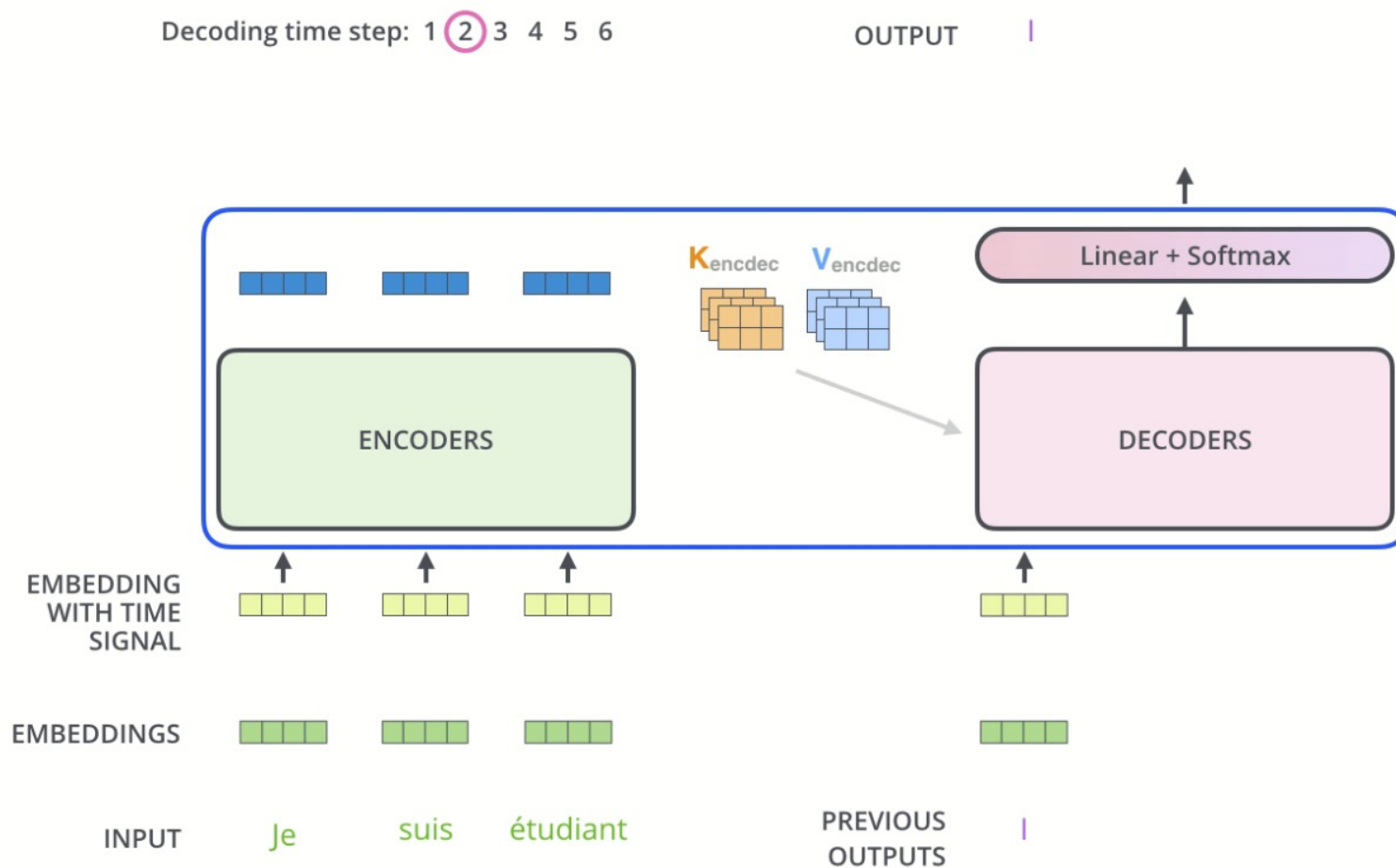
Transformer

• 解码器(decoder):



Transformer

- 解码器(decoder):

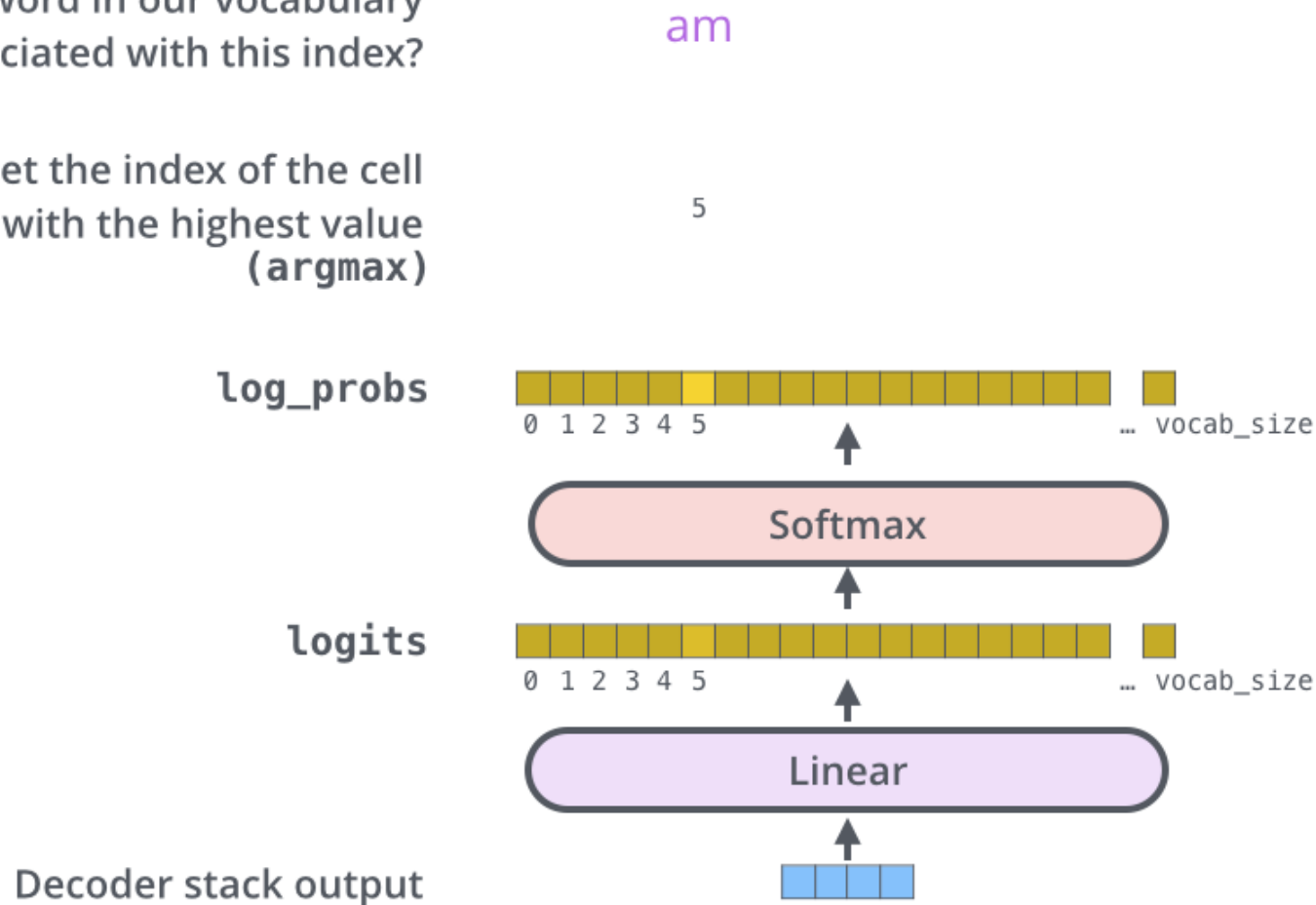


Transformer

- **解码器(decoder):** 最终层与Softmax层

Which word in our vocabulary
is associated with this index?

Get the index of the cell
with the highest value
(argmax)



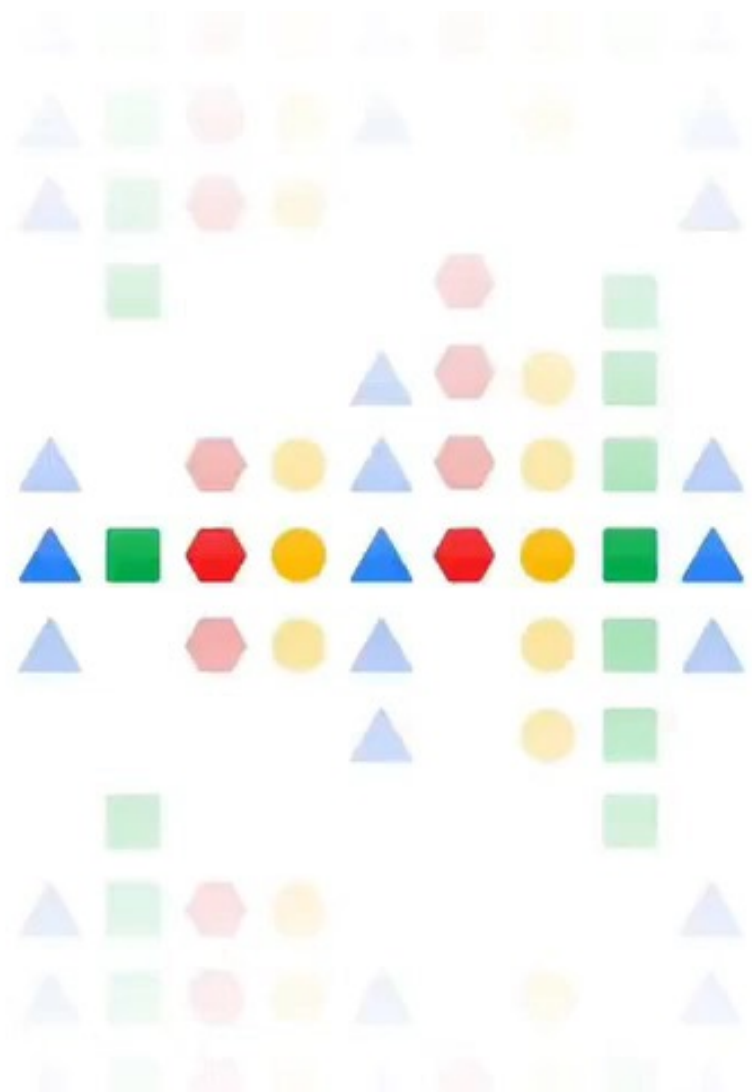
大语言模型概览



北京大学
PEKING UNIVERSITY

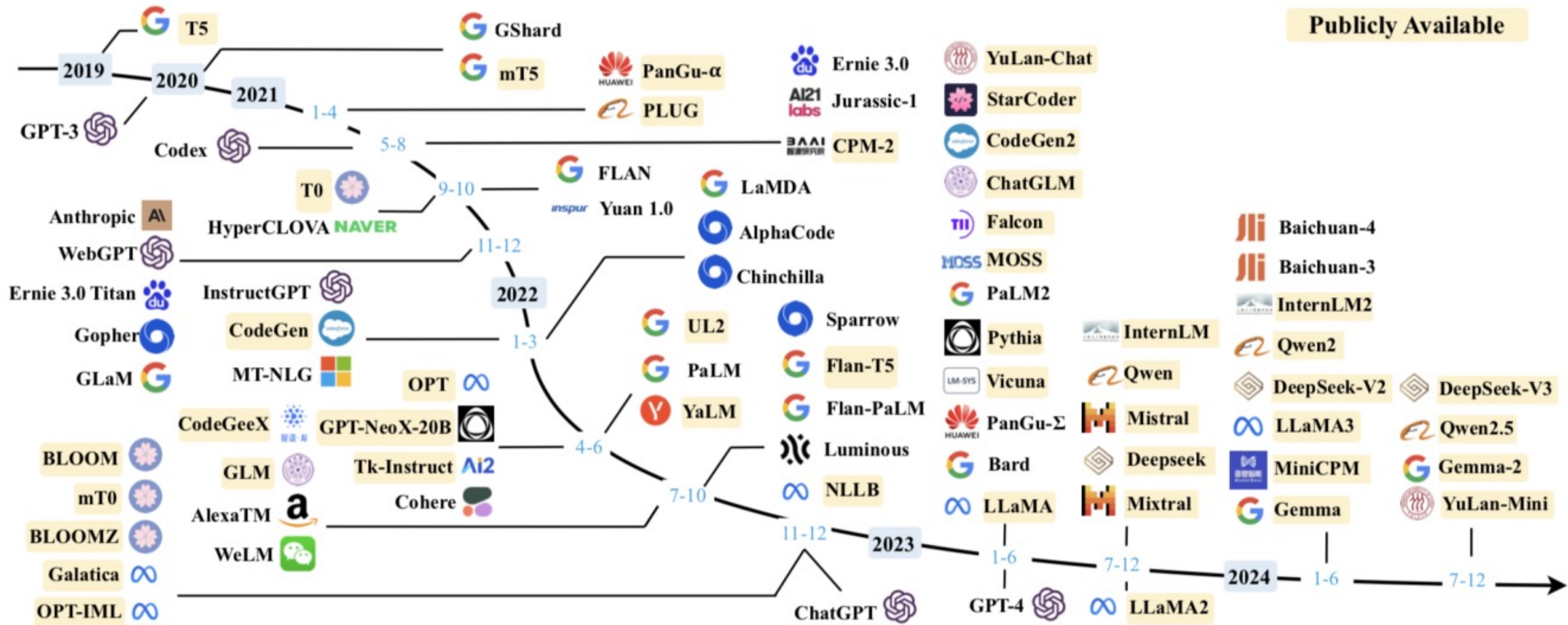
Google Cloud

Introduction to Large Language Models



https://www.cloudskillsboost.google/course_templates/539/video/414765

大语言模型概览



大语言模型概览

Rank	Model	Elo Rating
1	 vicuna-13b	1169
2	 koala-13b	1082
3	 oasst-pythia-12b	1065
4	alpaca-13b	1008
5	chatglm-6b	985
6	fastchat-t5-3b	951
7	dolly-v2-12b	944
8	llama-13b	932
9	stablelm-tuned-alpha-7b	858

Model A

Correct grammar: I are happy.

I'm happy to help you with your English grammar. Can you please provide me with a sentence or two that you would like me to correct?

A is better

B is better

Tie

Both are bad

Model B

Correct grammar: I are happy.

The correct grammar would be "I am happy."

A is better

B is better

Tie

Both are bad

国内排名最高的开源大语言模型（2023年5月）

	HELM / lm-evaluation-harness	OpenAI/eval	Alpaca Evaluation	Vicuna Evaluation	Chatbot Arena
Question Source	Academic datasets	Mixed	Self-instruct evaluation set	GPT-4 generated	User prompts
Evaluator	Program	Program/Model	Human	GPT-4	User
Metrics	Basic metrics	Basic metrics	Win rate	Win rate	Elo ratings

大语言模型概览



北京大学
PEKING UNIVERSITY

Rank* (UB)	Model	Arena Elo	95% CI	Votes	Organization	License	Knowledge Cutoff
1	GPT-4-Turbo-2024-04-09	1258	+3/-3	44592	OpenAI	Proprietary	2023/12
2	GPT-4-1106-preview	1252	+2/-3	76173	OpenAI	Proprietary	2023/4
2	Gemini 1.5 Pro API-0409-Preview	1249	+3/-3	61011	Google	Proprietary	2023/11
2	Claude 3 Opus	1248	+2/-2	101063	Anthropic	Proprietary	2023/8
3	GPT-4-0125-preview	1246	+3/-2	70239	OpenAI	Proprietary	2023/12
6	Bard (Gemini Pro)	1208	+5/-6	12387	Google	Proprietary	Online
6	Llama-3-70b-Instruct	1208	+3/-3	75844	Meta	Llama 3 Community	2023/12
7	Reka-Core-20240501	1199	+4/-4	18735	Reka AI	Proprietary	Unknown
8	Claude 3 Sonnet	1200	+2/-3	84252	Anthropic	Proprietary	2023/8
10	GPT-4-0314	1189	+3/-3	53446	OpenAI	Proprietary	2021/9
10	Qwen-Max-0428	1186	+5/-7	10508	Alibaba	Proprietary	Unknown
10	Command R+	1189	+3/-3	50490	Cohere	CC-BY-NC-4.0	2024/3
12	Claude 3 Haiku	1180	+2/-3	74897	Anthropic	Proprietary	2023/8
13	Qwen1.5-110B-Chat	1172	+7/-8	6019	Alibaba	Qianwen LICENSE	2024/4
14	GPT-4-0613	1165	+3/-3	73295	OpenAI	Proprietary	2021/9

国内排名最高的大语言模型
(2024年5月)

<https://chat.lmsys.org/?leaderboard>

大语言模型概览



北京大学
PEKING UNIVERSITY

Rank* (UB)	Rank (StyleCtrl)	Model	Arena Score	95% CI	Votes	Organization	License
1	1	Gemini-2.5-Pro-Exp-03-25	1439	+7/-5	9013	Google	Proprietary
2	2	ChatGPT-4o-latest (2025-03-26)	1407	+6/-6	8261	OpenAI	Proprietary
2	4	Grok-3-Preview-02-24	1402	+5/-3	14849	xAI	Proprietary
2	2	GPT-4.5-Preview	1398	+5/-6	14520	OpenAI	Proprietary
2	4	Gemini-2.5-Flash-Preview-04-17	1392	+10/-13	3325	Google	Proprietary
5	9	Gemini-2.0-Flash-Thinking-Exp-01-21	1380	+4/-3	26309	Google	Proprietary
5	5	Gemini-2.0-Pro-Exp-02-05	1380	+5/-4	20127	Google	Proprietary
5	4	DeepSeek-V3-0324	1372	+9/-6	5888	DeepSeek	MIT
9	6	DeepSeek-R1	1358	+4/-5	16077	DeepSeek	MIT
9	14	Gemini-2.0-Flash-001	1354	+3/-4	22201	Google	Proprietary
10	6	o1-2024-12-17	1350	+3/-3	28990	OpenAI	Proprietary
11	14	Gemma-3-27B-it	1342	+5/-6	9976	Google	Gemma
12	14	Qwen2.5-Max	1340	+4/-4	21169	Alibaba	Proprietary
12	11	o1-preview	1335	+3/-3	33170	OpenAI	Proprietary
15	14	o3-mini-high	1325	+4/-4	17988	OpenAI	Proprietary
15	17	DeepSeek-V3	1318	+4/-3	22846	DeepSeek	DeepSeek
15	21	QwQ-32B	1316	+6/-6	7735	Alibaba	Apache 2.0

国内排名最高的大语言模型
(2025年4月)

<https://lmarena.ai/?leaderboard>

大语言模型概览



北京大学
PEKING UNIVERSITY

Rank ↕	Rank Spread ⓘ	Model ↕	Score ↓	Votes ↕	Price \$/M ⓘ ↕	Context ↕
1	1 - 7	claude-opus-4-6 Anthropic · Proprietary	1545 ±14	2,077	\$5 / \$25	1M
2	1 - 8	claude-opus-4-6-thinking Anthropic · Proprietary	1543 ±15	1,690	\$5 / \$25	1M
3	1 - 12	claude-opus-4-7-thinking Anthropic · Proprietary	1541 ±19	874	\$5 / \$25	1M
4	1 - 14	claude-opus-4-7 Anthropic · Proprietary	1536 ±18	1,006	\$5 / \$25	1M
5	1 - 26	gpt-5.5-high OpenAI · Proprietary	1534 ±24	602	\$5 / \$30	1.1M
6	1 - 27	gpt-5.4-high OpenAI · Proprietary	1524 ±16	1,404	\$2.50 / \$15	1.1M
7	2 - 29	gemini-3.1-pro-preview Google · Proprietary	1518 ±12	2,408	\$2 / \$12	1M
8	1 - 43	mimo-v2.5-pro Xiaomi · MIT	1514 ±24	553	\$1 / \$3	1M
9	3 - 42	qwen3.5-max-preview Alibaba · Proprietary	1507 ±16	1,259	N/A	N/A
10	4 - 44	claude-sonnet-4-6 Anthropic · Proprietary	1505 ±16	1,442	\$3 / \$15	1M

国内排名最高的
大语言模型
(2026年4月)

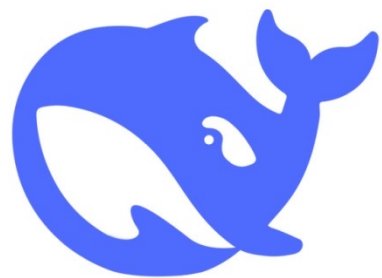
<https://lmarena.ai/?leaderboard>

大语言模型概览-code

1	1 ~ 2	 claude-opus-4-7-thinking Anthropic · Proprietary	1570 +12/-12	3,309	\$5 / \$25	1M
2	1 ~ 4	 claude-opus-4-7 Anthropic · Proprietary	1560 +12/-12	3,333	\$5 / \$25	1M
3	2 ~ 5	 claude-opus-4-6-thinking Anthropic · Proprietary	1549 +9/-9	6,382	\$5 / \$25	1M
4	2 ~ 5	 claude-opus-4-6 Anthropic · Proprietary	1544 +8/-8	7,325	\$5 / \$25	1M
5	3 ~ 8	 glm-5.1 Z.ai · MIT	1531 +11/-11	3,609	\$1.40 / \$4.40	202.8K
6	5 ~ 8	 claude-sonnet-4-6 Anthropic · Proprietary	1524 +8/-8	9,434	\$3 / \$15	1M
7	5 ~ 8	 kimi-k2.6 Moonshot · Modified MIT	1523 +12/-12	2,651	\$0.95 / \$4	262.1K
8	5 ~ 11	 muse-spark Meta · Proprietary	1509 +16/-16  Preliminary	1,629	N/A	N/A
9	8 ~ 12	 gpt-5.5-high (codex-harness) OpenAI · Proprietary	1491 +12/-12	2,765	N/A	N/A
10	8 ~ 11	 claude-opus-4-5-20251101-thinking-32k Anthropic · Proprietary	1490 +7/-7	13,063	\$5 / \$25	200K
11	8 ~ 16	 qwen3.6-max-preview Alibaba · Proprietary	1478 +17/-17	1,343	\$1.04 / \$6.24	262.1K
12	10 ~ 16	 mimo-v2.5-pro Xiaomi · MIT	1472 +11/-11	3,278	\$1 / \$3	1M

国内排名最高的
大语言模型
(2026年4月)

DeepSeek



deepseek



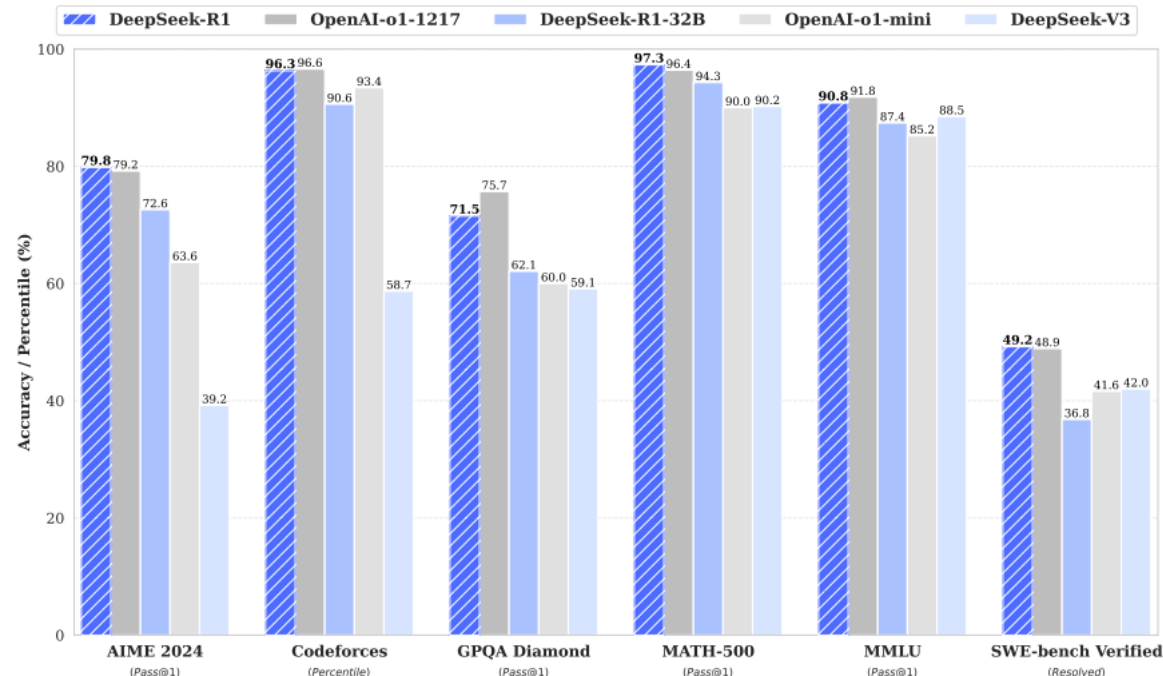
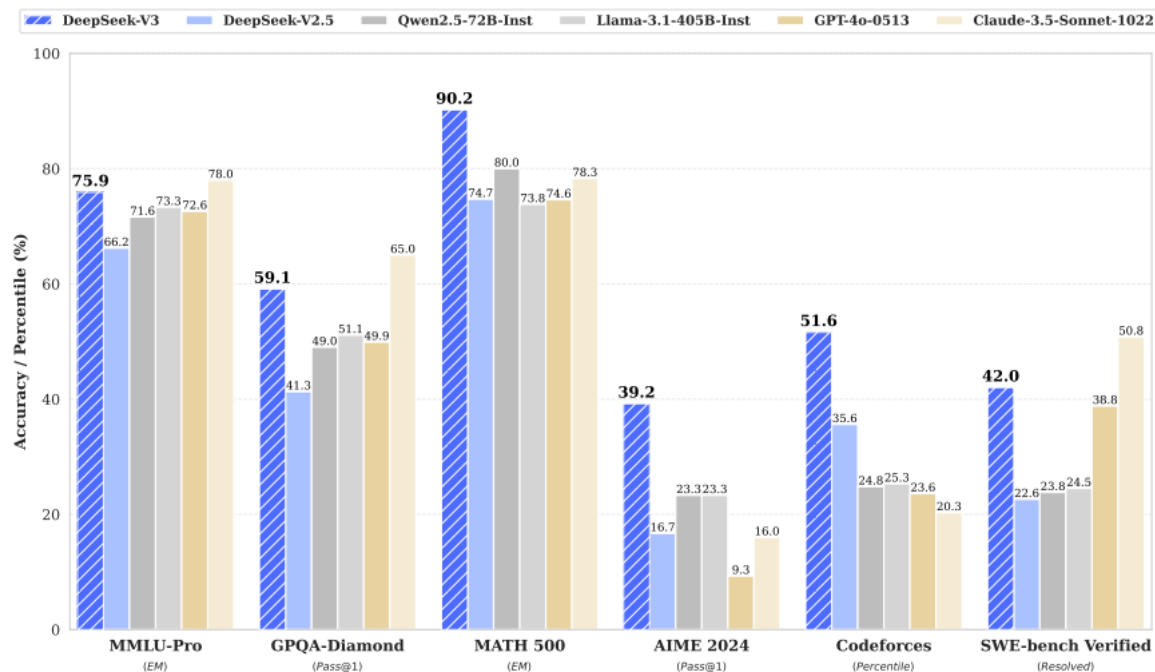
北京大学
PEKING UNIVERSITY

- ❑ **2025年1月下旬**，DeepSeek的R1模型发布后的一周内，DeepSeek刷屏美国各大主流媒体和社交网站。其中一部分原因为，TMT Breakout在与网友的讨论中，隐隐将英伟达周五下跌的原因指向DeepSeek的爆火。即R1的成功可能削弱了市场对英伟达AI芯片需求的预期，导致交易员做空英伟达股票，进而引发股价下跌。
- ❑ **2025年1月22日**，美国媒体Business Insider报道称，DeepSeek-R1模型秉承开放精神，完全开源，为美国AI玩家带来了麻烦。开源的先进AI可能挑战那些试图通过出售技术赚取巨额利润的公司。
- ❑ **2025年1月26日**，有网友反应，DeepSeek崩了，提示服务器繁忙。新浪科技询问DeepSeek今天下午是否有闪崩时，DeepSeek回应称：1月26日下午DeepSeek确实出现了局部服务波动，但问题在数分钟内得到解决。此次事件可能与新模型发布后的访问量激增有关，而官方状态页未将其标记为事故。
- ❑ **2025年1月27日**，来自国产大模型公司“深度求索”的DeepSeek应用登顶苹果美国地区应用商店免费APP下载排行榜，在美区下载榜上超越了ChatGPT。同日，苹果中国区应用商店免费榜显示，DeepSeek成为中国区第一。

DeepSeek



北京大学
PEKING UNIVERSITY



HAI-LLM	DeepSeek LLM	DeepSeek Coder	DeepSeek Math	DeepSeek VL	DeepSeek V2	DeepSeek Coder-V2	DeepSeek V3	DeepSeek R1
2023.6	2024.1	2024.1	2024.2	2024.3	2024.5	2024.6	2024.12	2025.1
- 深度学习训练框架，支持万亿参数、数千GPU训练 - 多种并行策略	- 细化扩展定律（超参数） 7B、67B参数 2T词元 > GPT-3.5	16K窗口 - 代码相关数据收集清洗 1.3B-33B > GPT-3.5	- 基于7B Coder - 提出GRPO - 数学数据收集清洗，120B > GPT-3.5	- 图文模型，基于DeepSeek LLM - 图文数据收集 - 高分辨率 < GPT-4V (7B)	- MoE, 236B 128K 窗口 - MLA注意力 - 减少训练代价 ≈ Mixtral	16B、236B 128K 窗口 - 基于V2训练6T数据，共10.2T ≈ GPT-4 Turbo	- MoE, 671B - MLA, MTP - 关注训练infra与工程技术 ≈ GPT-4o	- 基于V3训练 - 长思维链模型 - Scaling RL - 探索蒸馏技术 ≈ o1

Training Costs	Pre-Training	Context Extension	Post-Training	Total
in H800 GPU Hours	2664K	119K	5K	2788K
in USD	\$5.328M	\$0.238M	\$0.01M	\$5.576M

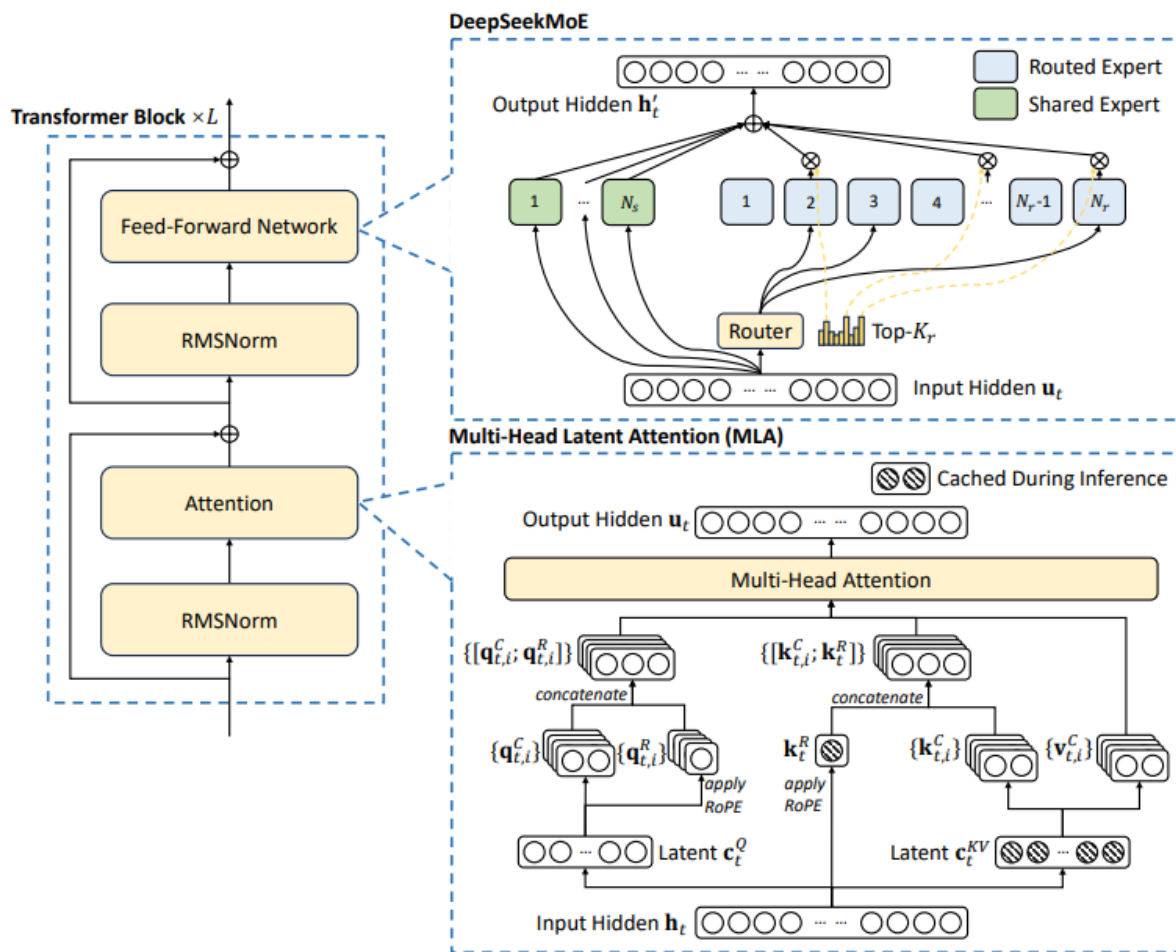
Table 1 | Training costs of DeepSeek-V3, assuming the rental price of H800 is \$2 per GPU hour.

Llama3.1-405B的预训练：20.7 百万 H100 卡时，至少是DeepSeek V3的7.8倍

为什么 DeepSeek 会引起世界关注？

- 模型开源、技术开放、免费商用
- 极致的算法与工程优化，显著降低算力需求
- 打破海外闭源产品的领先时效性
- 达到了与OpenAI现有API性能可比的水平

DeepSeek V3

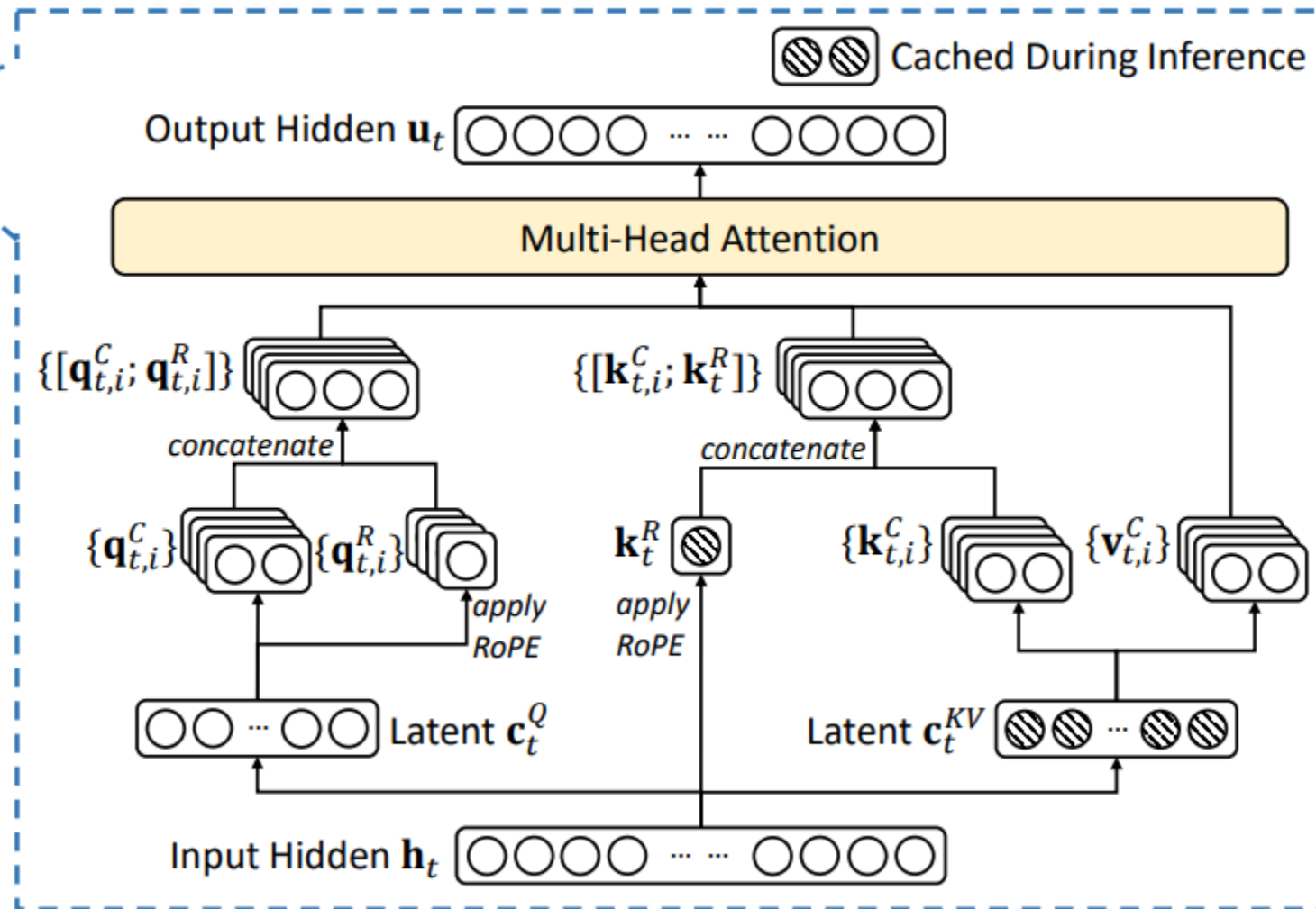


DeepSeek V3的核心技术创新:

- DeepSeekMoE:** 采用混合专家模型，61个Transformer层中各自包含256个专家和1个共享专家，每次激活8个专家，总参数671B，每次激活37B，V3 是第一个将MoE做到200+ 专家并训练稳定
- MLA:** 采用多头隐式注意力机制，引入低秩分解技术，可极大减小KV缓存开销(减少93.7%)、提高计算效率
- DualPipe:** 极致的流水线并行，通信-计算覆盖。实现1) 流水线气泡的大幅压缩；2) 将前向和后向计算与通讯阶段重叠减少计算资源闲置；3) 硬件资源的高效复用
- 参数量化等:** 例如，采用FP8乘法使得显存下降75%，指令吞吐量提升4倍
- MTP:** 传统自回归模型逐个预测token，而MTP在单次前向传播中可预测多个未来token，生成效率成倍提升

DeepSeek V3 - MLA

Multi-Head Latent Attention (MLA)



MLA的核心思路：对注意力机制中的K和V进行低秩联合压缩，以在推理过程中减少KV的缓存需求。

$$\mathbf{c}_t^{KV} = W^{DKV} \mathbf{h}_t,$$

$$[\mathbf{k}_{t,1}^C; \mathbf{k}_{t,2}^C; \dots; \mathbf{k}_{t,n_h}^C] = \mathbf{k}_t^C = W^{UK} \mathbf{c}_t^{KV},$$

$$\mathbf{k}_t^R = \text{RoPE}(W^{KR} \mathbf{h}_t),$$

$$\mathbf{k}_{t,i} = [\mathbf{k}_{t,i}^C; \mathbf{k}_t^R],$$

$$[\mathbf{v}_{t,1}^C; \mathbf{v}_{t,2}^C; \dots; \mathbf{v}_{t,n_h}^C] = \mathbf{v}_t^C = W^{UV} \mathbf{c}_t^{KV},$$

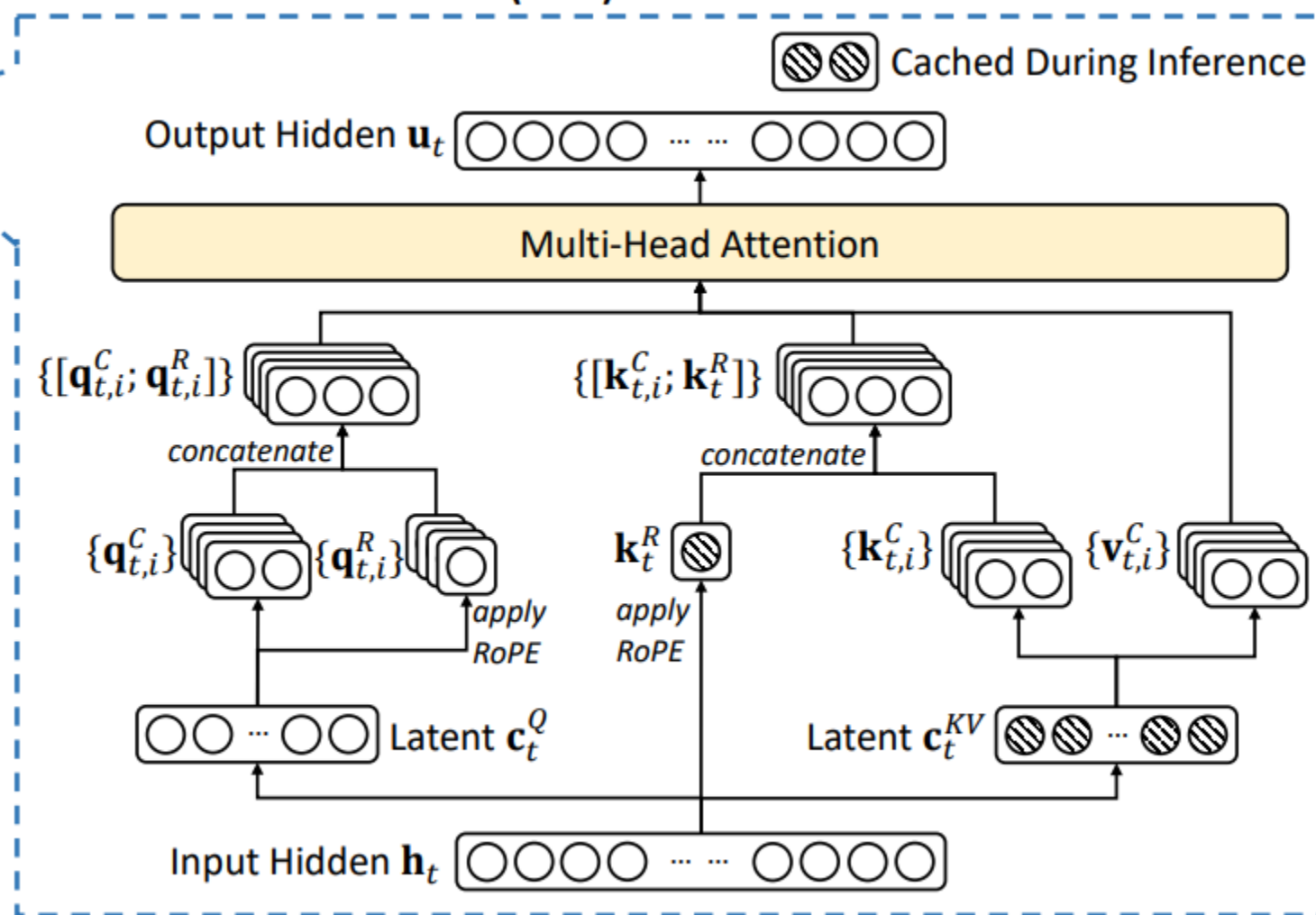
RoPE：旋转位置编码，“通过绝对位置编码的方式实现相对位置编码”

$$\mathbf{q}_m^\top \mathbf{k}_n = (\mathbf{R}_{\Theta,m}^d \mathbf{W}_q \mathbf{x}_m)^\top (\mathbf{R}_{\Theta,n}^d \mathbf{W}_k \mathbf{x}_n) = \mathbf{x}^\top \mathbf{W}_q \mathbf{R}_{\Theta,n-m}^d \mathbf{W}_k \mathbf{x}_n$$

$$\mathbf{R}_{\Theta,n-m}^d = (\mathbf{R}_{\Theta,m}^d)^\top \mathbf{R}_{\Theta,n}^d, \quad \mathbf{R}_{\Theta,m}^d = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \dots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \dots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \dots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix}$$

DeepSeek V3 - MLA

Multi-Head Latent Attention (MLA)



MLA的核心思路：对注意力机制中的K和V进行低秩联合压缩，以在推理过程中减少KV的缓存需求。

$$\begin{aligned}
 \mathbf{c}_t^Q &= W^{DQ} \mathbf{h}_t, \\
 [\mathbf{q}_{t,1}^C; \mathbf{q}_{t,2}^C; \dots; \mathbf{q}_{t,n_h}^C] &= \mathbf{q}_t^C = W^{UQ} \mathbf{c}_t^Q, \\
 [\mathbf{q}_{t,1}^R; \mathbf{q}_{t,2}^R; \dots; \mathbf{q}_{t,n_h}^R] &= \mathbf{q}_t^R = \text{RoPE}(W^{QR} \mathbf{c}_t^Q), \\
 \mathbf{q}_{t,i} &= [\mathbf{q}_{t,i}^C; \mathbf{q}_{t,i}^R],
 \end{aligned}$$

最终输出：

$$\begin{aligned}
 \mathbf{o}_{t,i} &= \sum_{j=1}^t \text{Softmax}_j \left(\frac{\mathbf{q}_{t,i}^T \mathbf{k}_{j,i}}{\sqrt{d_h + d_h^R}} \right) \mathbf{v}_{j,i}^C, \\
 \mathbf{u}_t &= W^O [\mathbf{o}_{t,1}; \mathbf{o}_{t,2}; \dots; \mathbf{o}_{t,n_h}],
 \end{aligned}$$

DeepSeek
V3

冷启动微调
(数千样本)

推理任务上
强化学习

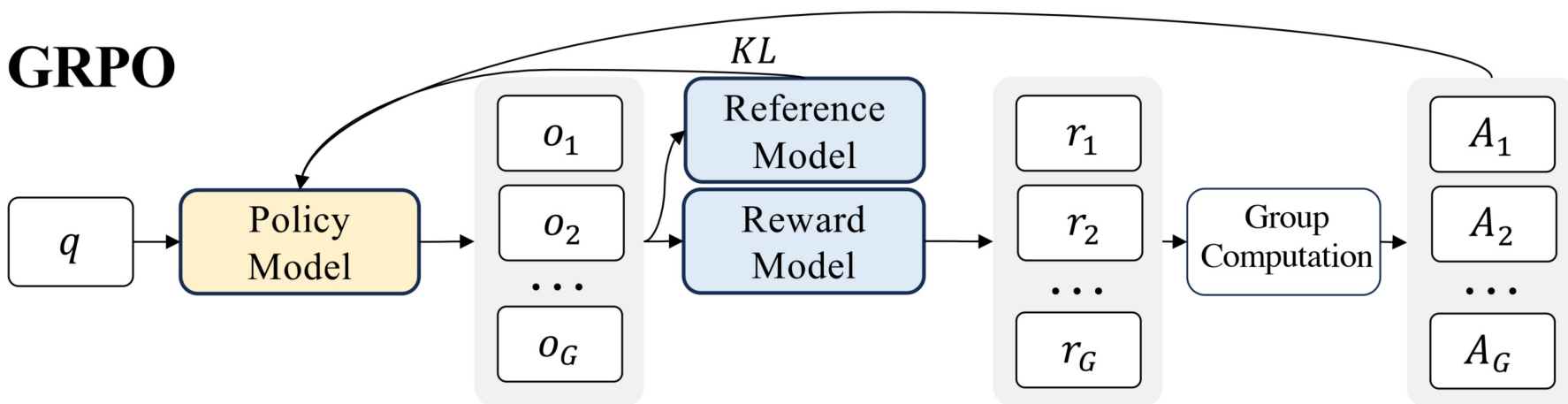
拒绝采样微调
(600K + 200K)

DeepSeek
R1

核心思路：冷启动SFT+RL多阶段训练

- 先用SFT赋予基础推理能力，再通过RL优化输出质量
- 在RL阶段引入组相对策略优化GRPO

GRPO



DeepSeek R1

直接使用筛选过的DeepSeek-R1生成的800K数据用来SFT蒸馏其他小模型

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
GPT-4o-0513	9.3	13.4	74.6	49.9	32.9	759
Claude-3.5-Sonnet-1022	16.0	26.7	78.3	65.0	38.9	717
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9	1316
DeepSeek-R1-Distill-Qwen-1.5B	28.9	52.7	83.9	33.8	16.9	954
DeepSeek-R1-Distill-Qwen-7B	55.5	83.3	92.8	49.1	37.6	1189
DeepSeek-R1-Distill-Qwen-14B	69.7	80.0	93.9	59.1	53.1	1481
DeepSeek-R1-Distill-Qwen-32B	72.6	83.3	94.3	62.1	57.2	1691
DeepSeek-R1-Distill-Llama-8B	50.4	80.0	89.1	49.0	39.6	1205
DeepSeek-R1-Distill-Llama-70B	70.0	86.7	94.5	65.2	57.5	1633

总结

Attention is all you need!



谢谢



北京大学
PEKING UNIVERSITY

